

# Long-Run Exchange Rate Modeling

## A Survey of the Recent Evidence

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*In this paper the recent literature on long-run exchange rate modeling is surveyed. In particular, we review the voluminous literature that tests for a unit root in real exchange rates and the closely related work on testing for a unit root in the residual from a regression of the nominal exchange rate on relative prices. We argue that the balance of evidence is supportive of the existence of some form of long-run exchange rate relationship. The form of this relationship, however, does not accord exactly with a traditional representation of the long-run exchange rate, and we offer some potential explanations. [JEL F31]*

IN THIS PAPER we survey the recent literature on long-run, or equilibrium, exchange rate modeling.<sup>1</sup> Although, of course, interest in the determinants of long-run exchange rates is not new, the topic has been revived by recent developments in the time-series literature, particularly those relating to cointegration and unit root testing. The tenor of the conclusions contained in this paper may be summarized in the following way. Ten years or so ago, the consensus view in the economics profession was that, from an empirical perspective, the long-run exchange rate was not well defined.<sup>2</sup> Today, the evidence summarized in this paper would lead one to the opposite conclusion. There is now overwhelming evidence to indicate that economists can say something positive about long-run

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<sup>1</sup> There are a number of surveys of the earlier literature on long-run modeling—see, for example, Frenkel (1981), Mussa (1979), and MacDonald (1988a). See also Breuer (1994) and Froot and Rogoff (1994) for alternative surveys of the recent literature on long-run exchange rate modeling.

<sup>2</sup> See, for instance, the discussion in Frenkel (1981) and Krugman (1978).

exchange rates. This is clearly a relief. Such long-run relationships relate either directly or indirectly (in the case of the monetary model) to some form of absolute purchasing power parity (PPP): the hypothesis that a long-run exchange rate is determined by domestic prices relative to foreign prices. The existence of empirically verifiable long-run exchange rate relationships provides a firm foundation on which to build what may be referred to as medium-run exchange rate models; that is, exchange rate models that capture exchange rate movements over the economic cycle. Recently such modeling has also gained a new lease on life, and we briefly note some of the relevant papers in our concluding section.

The paper has four main components. In Section I, we present an overview of the concepts of absolute and relative PPP and some general issues concerning their validity. Section II relates the concepts of absolute and efficient markets PPP (EMPPP) to a standard balance of payments equilibrium condition. Although this condition has become an unfashionable framework of late, we nevertheless believe it is insightful in the current context. Thus, it may be used to demonstrate the fact that absolute PPP emanates from goods arbitrage on the current account, and the strict conditions necessary for this to hold, while EMPPP is generated from arbitrage on the capital account. The use of the balance of payments condition is also helpful in distinguishing between the concepts of statistical and "true" economic equilibria (a distinction that is important when trying to interpret recent empirical work). Section III details recent empirical work on absolute PPP using cointegration methods; we demonstrate that there is now considerable support for some form of long-run PPP. The empirical evidence on the efficient markets view of PPP and, in particular, the time-series properties of the real exchange rate, is discussed in Section IV. We demonstrate in Section IV that the real exchange rate is a mean-reverting series and this may be interpreted as further evidence in favor of absolute PPP, but unfavorable to efficient markets PPP. Also considered in Section IV are a number of papers that seek to model the real exchange rate. The paper closes with some concluding comments, which incorporate a discussion of how the findings of this paper relate to the rapidly growing literature on dynamic exchange rate modeling. In particular, this involves a discussion of how to define a sensible long-run exchange rate relationship using only data from the recent floating period.

## **I. Absolute and Relative Purchasing Power Parity**

The condition of absolute PPP is usually derived in a two-country setting in which the home and foreign country each produce a range of

homogeneous tradeable goods (by which we mean a U.S.-produced refrigerator is identical to an Italian-produced refrigerator), the "law of one price" holding for each of the goods:

$$p_i^i = S_i p_i^{i*}, \quad (1)$$

where  $p_i^i$  denotes the domestic price of good  $i$ ,  $S_i$  denotes the home currency price of a unit of foreign currency, and an asterisk denotes a foreign magnitude. Condition (1) is maintained by arbitrage. Thus, if for some reason the left-hand side of (1) is greater than the right-hand side, it would be profitable to ship the good from the foreign country to the domestic country thereby forcing the domestic currency value of the foreign good up (by a rise in  $S_i$  and/or  $p_i^{i*}$ ) and the domestic price of the good down, until equality between the two prices is restored. By summing the prices of all of the traded goods in each country, and giving each price the same weight in the sum, we obtain the condition of absolute PPP.

$$S_i = P_i / P_i^*, \quad (2)$$

where  $P_i = \sum_{i=1}^n \alpha^i p_i^i$ ,  $P_i^* = \sum_{i=1}^n \alpha^i p_i^{i*}$ , and  $\alpha$  denotes a weight. An increase in the domestic price level, generated, say, by a monetary expansion, should result in an equiproportionate depreciation of the exchange rate. The restrictiveness of the absolute PPP hypothesis is clear: even if it were possible to construct prices in the manner suggested by condition (2) (we return to this point below), the existence of transportation costs and other impediments to trade, such as tariffs and quotas, will prevent condition (1) or (2) from holding exactly. However, if such factors are assumed constant over time, then either condition (1) or (2) would be expected to hold up to a constant factor  $\Pi$ .<sup>3</sup>

$$S_i = \Pi P_i / P_i^*, \quad (2')$$

or in logs

$$s_i = \pi + p_i - p_i^*, \quad (3)$$

where lowercase letters now indicate that the level of the variable has been transformed using the natural logarithm operator. On expressing the terms in equation (3) in changes we may obtain a weaker version of PPP, which is usually labeled relative PPP

$$\Delta s_i = \Delta p_i - \Delta p_i^*, \quad (4)$$

which states that the percentage exchange rate depreciation is equal to the difference between domestic and foreign inflation. Note that

<sup>3</sup>This, of course, still relies on the efficient functioning of goods markets. The factor  $\Pi$  could also incorporate constant differences in weights across countries.

equation (4) may be rearranged to produce an expression for the change in the real exchange rate: conditional on relative PPP holding, the logarithmic real exchange rate change should equal zero.

Often a proponent of PPP is understood as someone who believes that expressions like (2) and (4) hold continuously and at all times. However, it is clear from the writings of Cassel, and other prominent proponents of PPP, that the concept is often taken to be the level to which an exchange rate gravitates. Thus there are seen to be an array of factors, such as central bank intervention and long- and short-term capital flows (see Officer (1976) for a review of these factors), that keep the actual exchange rate away from its PPP-determined rate. Eventually, however, the exchange rate should move in line with its PPP rate. To use a time-series expression, discussed in some detail later, the real exchange rate under this view is mean reverting. We label this view of PPP the Cassellian view. The distinction will be useful below when we come to discuss the efficient markets approach to PPP.

The last point also raises the issue of causation. In both of the concepts of PPP discussed above, causation supposedly runs from prices to the exchange rate. However, in circumstances in which we have short-run real exchange rate changes, it is possible for causation to run in the opposite direction. For example, consider the situation in which from a position of absolute PPP holding there is a one-shot capital outflow from the domestic country, thereby depreciating the nominal rate and, with sticky prices in the short run, the real rate. In the Cassellian view this change in the real rate should be corrected, but how? It is clearly possible that this adjustment takes place, at least in part if not wholly, by prices reacting to the initial exchange rate change—indeed, this is essentially the story portrayed in the seminal article of Dornbusch (1976). This reverse causation, and joint endogeneity of exchange rates and prices, is especially likely to be a feature of actual data from the recent floating period.

In attempting to test either absolute or relative PPP, a researcher is immediately confronted with the issue of the appropriate price series to use. If one could construct price series consisting of the prices of homogeneous internationally traded goods, testing PPP would be relatively clean and straightforward. However, in practice such price measures are not available and researchers usually use consumer or wholesale price series.<sup>4</sup> Since both of these measures incorporate prices of nontraded goods, it is unlikely that their use in an empirical test would produce the symmetry and proportionality implied by conditions (2) and (4), although

<sup>4</sup>We do not consider recent research on the law of one price in this paper (the results are similar to those using aggregate price series).

these conditions are perhaps more likely to hold for tests constructed using wholesale prices, a series that contains a relatively large traded goods element. There are a number of other well-known problems that occur in trying to test PPP, and we discuss these in more detail in the next section.

It is worth noting that even if there are substantial nontraded elements in the price series used in an empirical test, relative PPP may still hold if the overall prices are homogeneous of degree one in monetary impulses. Thus, the so-called homogeneity postulate suggests that an increase in the money supply should leave equilibrium relative prices unchanged and should increase all prices by the same amount (this is discussed in more detail in Section III).

Throughout the paper we illustrate some of the key tests that have been employed in the recent literature using a standardized data base. In common with other researchers, we use a data set consisting of bilateral U.S. exchange rates and relative consumer and wholesale prices for nine currencies, over the period March 1973 to December 1992.<sup>5</sup> Given the points made above we must sound a cautionary note regarding the usefulness of such data in testing PPP. However, we believe that such tests are useful for illustrative purposes. Further, implementing a selection of tests on a unified data set should help to clarify whether or not the range of results reported in the literature, using different sample periods and different estimation methods, are sample specific or a function of the estimator used.

In Figures 1–9, we present plots of our group of exchange rates and relative domestic-U.S. consumer prices (the wholesale price plots are similar and are therefore not reported). For most of the currencies we note that there are lengthy periods in which an exchange rate tracks relative prices reasonably well and, in particular, there are often relatively long periods in which the trend behaviors of the two series are very similar. However, there are also currencies (for example, the U.K. pound sterling) and periods (for example, post-1986) for which the correspondence does not appear close.

A further common feature of these plots is that the variability of the exchange rates appears greater than that of relative prices. This stylized fact has led some commentators to argue that there is more to exchange rates than relative prices. This is certainly true in the short run—on a month to month, quarter to quarter basis—but is it equally true in the long run? That is the topic of this paper. In particular, we consider

<sup>5</sup>The data are those used in MacDonald and Moore (1994), the majority of which were extracted from the IMF's *International Financial Statistics*.

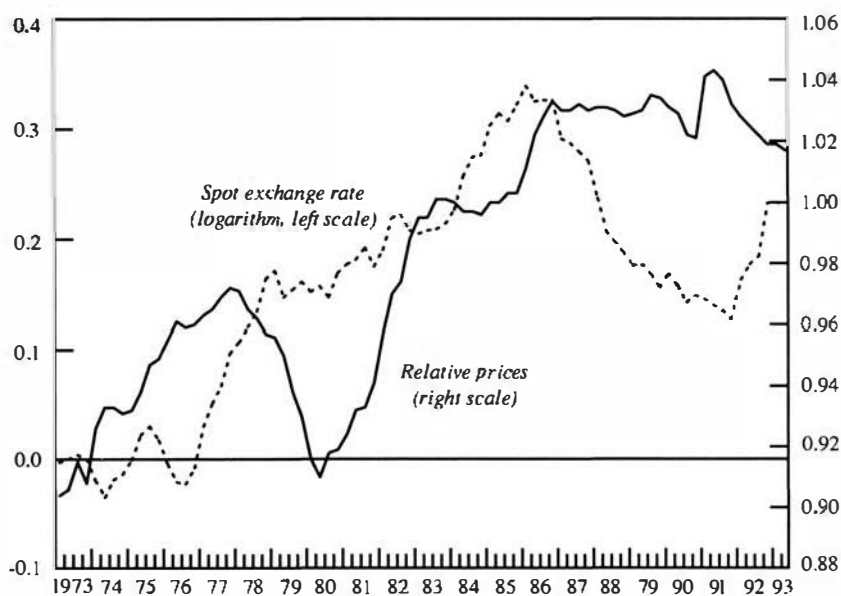
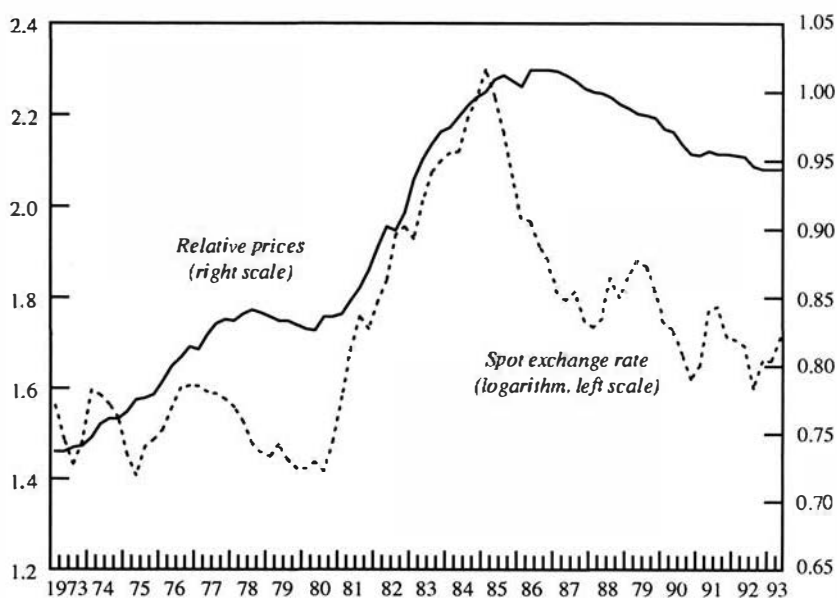
Figure 1. *Canadian Dollar – U.S. Dollar*Figure 2. *French Franc – U.S. Dollar*

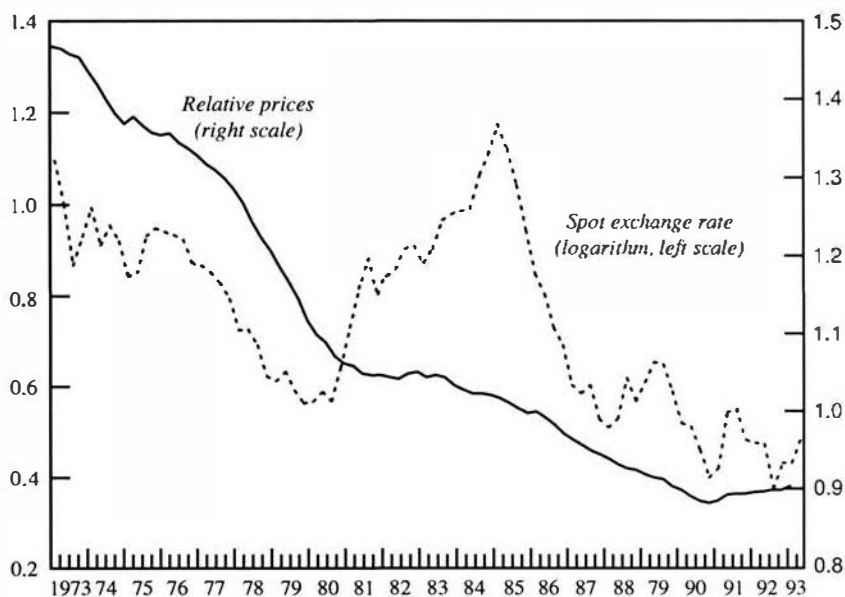
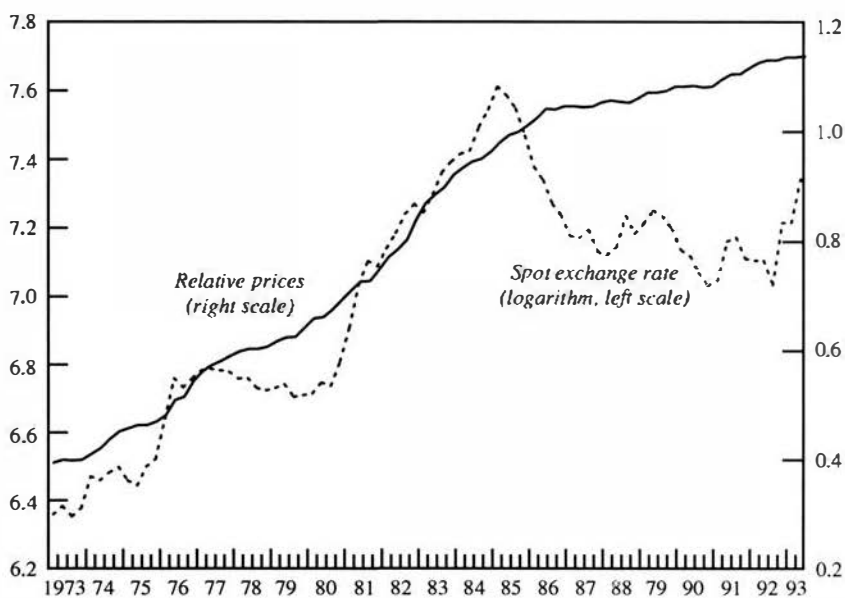
Figure 3. *German Mark – U.S. Dollar*Figure 4. *Italian Lira – U.S. Dollar*

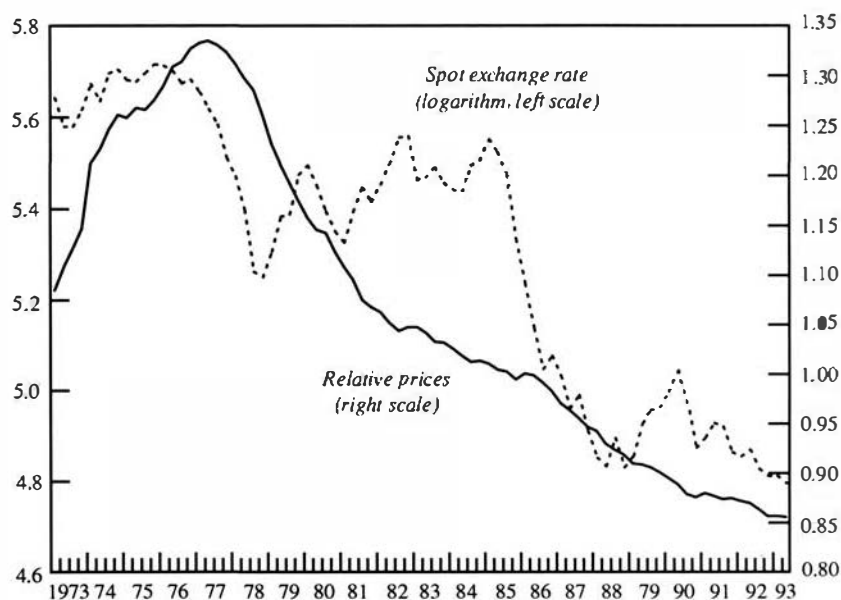
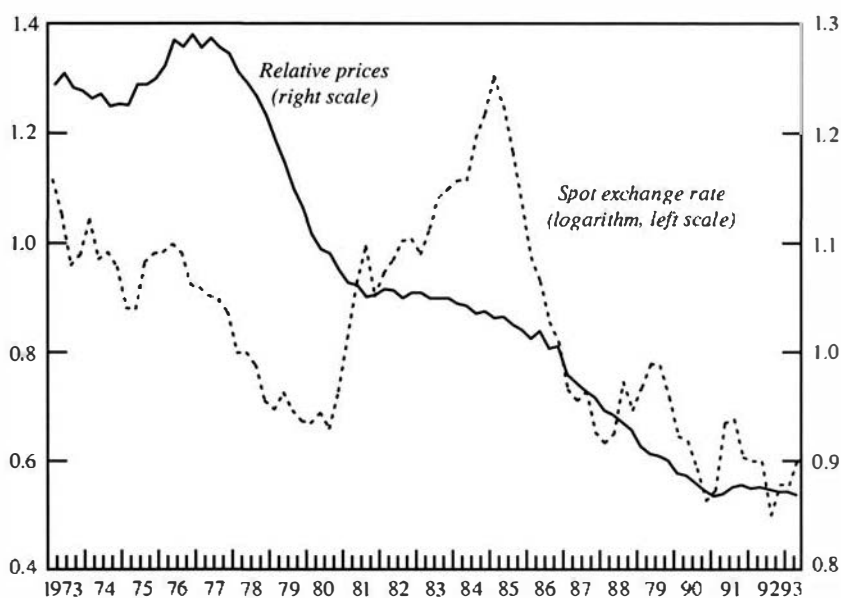
Figure 5. *Japanese Yen – U.S. Dollar*Figure 6. *Netherlands Guilder – U.S. Dollar*



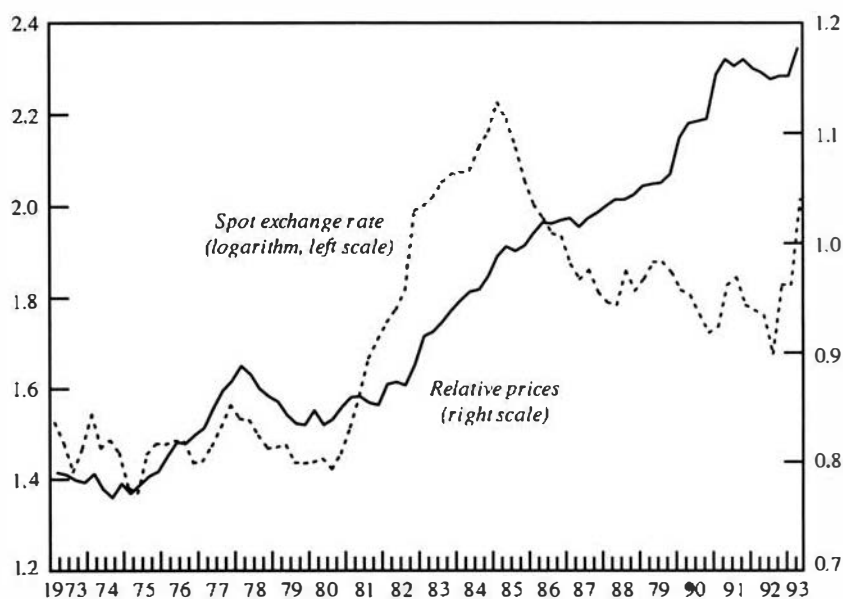
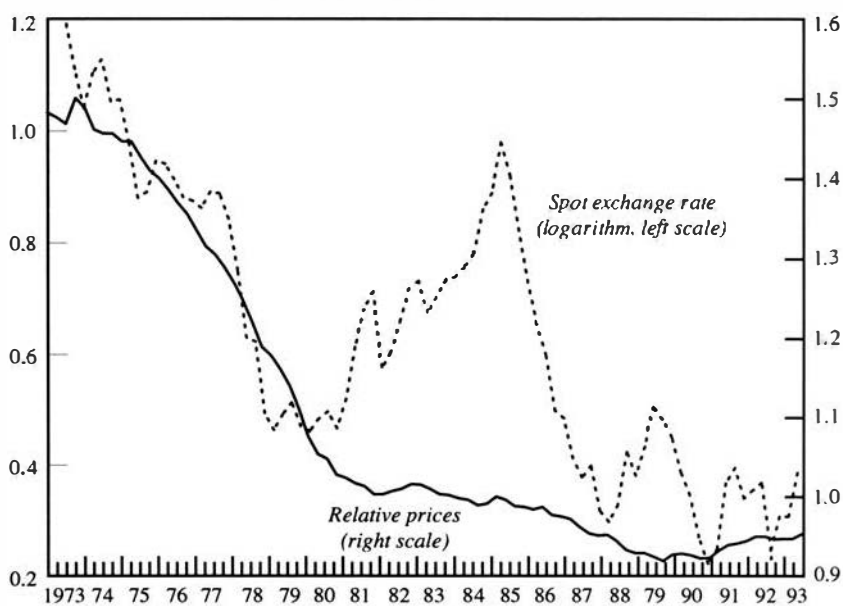
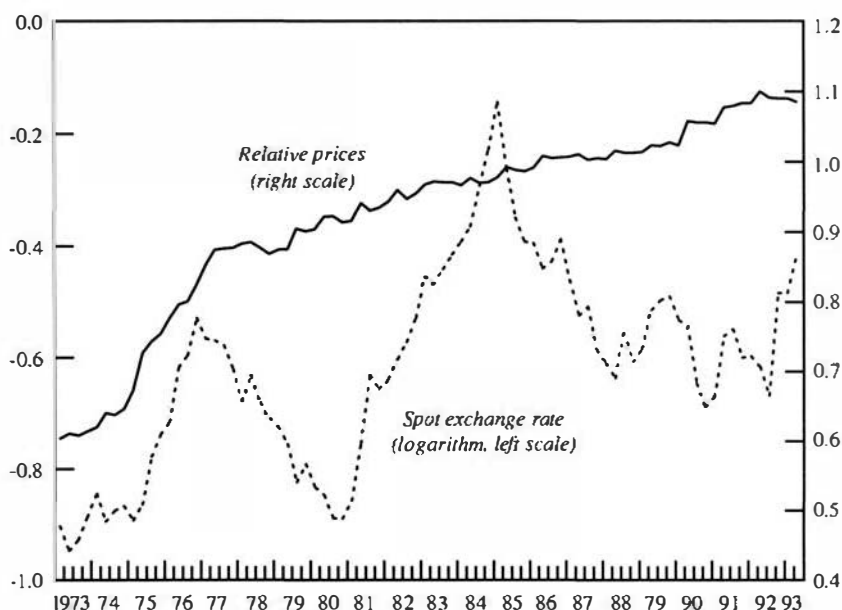
Figure 7. *Swedish Krona – U.S. Dollar*Figure 8. *Swiss Franc – U.S. Dollar*

Figure 9. Pound Sterling – U.S. Dollar



whether the large array of sophisticated econometric tests that have been conducted on the exchange rate/relative price relationship reveal more than our simple visual inspection of the data.

## II. The Balance of Payments and Purchasing Power Parity

A useful focal point for our discussion of long-run exchange rate modeling is the familiar balance of payments equilibrium condition under floating exchange rates:

$$ca_t + cap_t = \Delta f_t = 0, \quad (5)$$

where  $ca_t$  denotes the current account balance,  $cap_t$  denotes the capital account balance, and  $\Delta f_t$  denotes the change in reserves. Under freely floating exchange rates, the conventional balance of payments view of the determination of the exchange rate suggests that the exchange rate moves to equilibrate the sum of the current and capital accounts of the balance of payments, thereby ensuring that the change in reserves equals zero (see MacDonald (1988a)). A model in which balance of payments equilibrium is linked to conditions in asset markets, and which shows how the

exchange rate moves to ensure both stock and flow market equilibrium (where the latter is consistent with desired magnitudes), has been developed by Mussa (1984) and Frenkel and Mussa (1985).<sup>6</sup> There are two main reasons for using condition (5) as the focal point of our discussion. First, tests of absolute purchasing power parity can be shown to be related in a straightforward way to the current account, while tests focusing on the real exchange rate take as their starting point the capital account.<sup>7</sup> Second, the use of condition (5) also illustrates the ways in which simple PPP models, and other (related) real exchange relationships, may be deficient, and offers an appropriate way of testing PPP. The following equations summarize the assumptions regarding the current account:<sup>8</sup>

$$ca_t = nx_t + i_t^* A_t, \quad (6)$$

$$nx_t = \alpha(s_t + p_t^* - p_t) + \beta z_t; \quad \alpha > 0, \beta = ?, \quad (7)$$

where, of terms not previously defined,  $nx_t$  denotes net exports,  $A_t$  is the stock of net foreign assets,  $z_t$  captures exogenous influences on net exports,  $i_t^*$  denotes the foreign interest rate, and lowercase letters indicate that the level of the variable has been transformed using the natural logarithm operator.

Equation (7) indicates that net exports are dependent on the real exchange rate, or competitiveness, and on exogenous factors. For purposes of exposition, we have assumed that a country's competitiveness is a function of an overall price index, such as the CPI, which includes both traded and nontraded goods. The parameter  $\alpha$  is the elasticity of net exports with respect to competitiveness (see discussion below). The variable  $z_t$  is an exogenous catch-all term that captures, for example, expenditure effects from government and private consumers and productivity differences in the manufacture of traded goods between the home and foreign country. On substituting equations (6) and (7) into (5), and solving for the exchange rate we obtain

$$s_t = p_t - p_t^* - (\beta/\alpha)z_t - (i_t^*/\alpha)A_t - (1/\alpha)cap_t, \quad (8)$$

<sup>6</sup>The empirical implications of this model have been investigated by Faruquee (1995) and MacDonald (1995a).

<sup>7</sup>Not all researchers motivate their PPP modeling in quite this way. However, we believe that no great injustice is done by using this framework: the expositional advantages outweigh any disadvantage of presenting each researcher's account using his or her, perhaps slightly different, framework.

<sup>8</sup>Since the exchange rates considered in this paper are all bilateral rates, the implicit assumption in our discussion here is that the balance of payments accounts are defined on a bilateral basis, rather than on the conventional multilateral basis.

which may be thought of as a reduced form balance of payments equation for the exchange rate. Equation (8) is useful for motivating the two versions of PPP that have been widely tested in the recent exchange rate literature, namely, traditional absolute PPP and the EMPPP.

### Absolute PPP and the Current Account

What then is the process that allows us to move from expression (8) to the absolute PPP condition (2), re-expressed here in logs

$$s_t = p_t - p_t^* . \quad (2')$$

First, in some time frame, which may be referred to as the long run, it would be expected that net capital flows are zero (perhaps because net savings are at their desired level) and therefore the last term in equation (8) goes to zero; balance of payments equilibrium reduces to current account balance (this is the kind of stock-flow equilibrium captured in the portfolio balance model—see MacDonald (1988a)). Focusing on the current account items in equation (8), we note that strict absolute PPP requires  $z_t$  and  $A_t$  to be zero. One way of obtaining this would be to assume that  $\alpha$ , the elasticity of net exports with respect to relative prices, is infinitely large. This assumption is often made in textbook expositions of PPP; however, it has no empirical support (see Goldstein and Khan (1985)).

Non-zero values of  $A_t$  and  $z_t$  will produce a real exchange rate configuration that is not equal to zero. Hence even with full long-run price flexibility, changes in net excess demands for domestic goods can alter the relative price of traded to nontraded goods and therefore the real exchange rate. Examples of this would be the well-known Balassa-Samuelson productivity bias or changes in government and/or private consumption in favor of, say, domestic goods (see Hallwood and MacDonald (1994)). This effect is likely to be most important when comparing countries at different stages of development, but less important for countries at a similar level of development.<sup>9</sup> However, even if  $z_t$  is zero, note that any net foreign asset accumulation that has occurred in the move from equilibrium to equilibrium will require a value of the nominal exchange rate that does not simply reflect relative prices. This is one of

<sup>9</sup> Although even here it may be important. For example, comparisons of countries within the European Community are unlikely to generate any strong biases, whereas comparisons of EC countries and the United States with Japan may well produce very marked productivity biases for the post-war period (see discussion on p. 461).

the insights of the portfolio balance approach to the exchange rate and what Isard (1978) refers to as the “knockout punch” to absolute PPP. It is worth noting at this stage that even in a long-run context, a researcher will face a standard omitted variable bias in estimating a long-run PPP relationship, to the extent that  $(i_t^*/\alpha)$ ,  $A_t$ , and  $z_t$  are not included in the estimating equation. This makes clear the restrictiveness of the assumptions necessary for the purest form of PPP, even as a long-run concept.

A final point to note is that even if  $\alpha$  were high, as long as the degree of capital mobility is also high, capital flows would be an important reason for the violation of absolute PPP—this is essentially the Cassellian view outlined earlier. The Cassellian view may be expressed more formally as positing although there may be disturbances that push the nominal exchange rate away from the relative price configuration; these will not be permanent and will eventually be offset in the long term. To use a time-series expression, the real exchange rate under the Cassellian view is mean reverting:

$$q_t = pq_{t-1} + \epsilon_t, \quad 0 < p < 1 \quad (9)$$

where  $q_t$  denotes the real exchange rate,  $(s + p^* - p)_t$ ,  $\epsilon_t$  is a random error term,<sup>10</sup> and  $p$  should lie in the interval between zero and unity. An alternative approach to defining PPP, which is actually diametrically opposite to the Cassellian view, asserts that  $p$  in equation (9) is unity: shocks to the real exchange rate are permanent. This view of PPP, which we label the efficient markets view of PPP, is attributable to Adler and Lehmann (1983), Roll (1979), and MacDonald (1985a and 1985b) and relies for its derivation purely on the capital account of the balance of payments.

### The Capital Account of the Balance of Payments and Efficient Markets PPP

The efficient markets view of PPP asserts that in a world of high or perfect capital mobility it is not goods arbitrage that matters for the relationship between an exchange rate and relative prices, but interest rate arbitrage. The concept, which gives a fundamentally different prediction for the behavior of the real exchange rate than absolute PPP, may be illustrated in the following way. If capital is assumed perfectly mobile, as in the Mundell-Fleming and Dornbusch (1976) models, this immediately focuses attention on the capital account of the balance of payments.

<sup>10</sup> To allow for heteroscedastic disturbances,  $\epsilon_t$  need not be iid.

and, in a sense, it becomes the tail wagging the current account dog. If one views the capital account as reflecting the desired actions of agents, as in the models of Mussa (1984) and Frenkel and Mussa (1985), such tail-wagging may not be all that unappealing.<sup>11</sup> The assumption of perfect capital mobility may be represented as<sup>12</sup>

$$i_t - i_t^* = \Delta s_{t+k}^e, \quad (10)$$

which is the condition of uncovered interest rate parity (UIP), and where  $i_t$  denotes a nominal domestic interest rate,  $\Delta$  is the first difference operator, and an  $e$  denotes a subjective expectation (other symbols have the same interpretation as previously indicated).<sup>13</sup>

The relative nominal interest rate term in equation (10) may be decomposed into a real and expected inflation component using the Fisher decomposition

$$i_t - i_t^* = (r_t^e - r_t^{*e}) + (\Delta p_{t+k}^e - \Delta p_{t+k}^{*e}), \quad (11)$$

where  $r_t$  denotes the real interest rate and  $\Delta p_{t+k}^e$  denotes the expected inflation rate. On using this in the UIP condition we obtain

$$(s - p + p^*)_t = -(r^e - r^{*e})_{t,t+k} + (s^e - p^e + p^{*e})_{t,t+k}, \quad (12)$$

or

$$q_t = -(r^e - r^{*e})_{t,t+k} + q_{t+k}^e. \quad (12')$$

Equation (12') states that the current real exchange rate is determined by the (negative of) the expected real interest differential and the expected real rate in period  $t+k$  (this is sometimes interpreted as the equilibrium rate; that is,  $q_{t+k}^e = \bar{q}$  where  $\bar{q}$  denotes the equilibrium real rate).

Expression (12) may be rearranged to give a measure of the evolution of the expected real exchange rate:

$$(\Delta s^e - \Delta p^e + \Delta p^{*e})_{t+k} = (r^e - r^{*e})_{t,t+k}, \quad (13)$$

or

$$\Delta q_{t+k}^e = (r^e - r^{*e})_{t,t+k}. \quad (14)$$

<sup>11</sup> Although it surely is unappealing in the context of models in which the stock-flow repercussions of capital account imbalances are ignored, such as in the original textbook Mundell-Fleming model.

<sup>12</sup> By assuming the *cap* term in equation (8) equals  $\mu(i_t - i_t^* - \Delta s_{t+k}^e)$ , where the parameter  $\mu$  captures the mobility of international capital, perfect capital mobility occurs when  $\mu \rightarrow \infty$ .

<sup>13</sup> See Engel (1994) for a useful survey of the UIP condition.

On further assuming that the subjective expectations in equation (13) are equal to their rational counterparts, we can obtain<sup>14</sup>

$$\Delta s_{t+k} - \Delta p_{t+k} + \Delta p_{t+k}^* = (E_t r_t - E_t r_t^*) + \omega_{t+k}, \quad (15)$$

where the left-hand side is simply the change in the real exchange rate, which is driven by a real interest differential and a random error term. This may be referred to as the baseline real exchange rate model, attributable to Roll (1979), Adler and Lehmann (1983), and MacDonald (1985a and 1985b). If it is assumed that real interest rates are exactly equalized across countries, we have

$$q_t - q_{t-1} = \omega_t. \quad (16)$$

This model can imply, therefore, that the real exchange rate follows a random walk.<sup>15</sup> It is worth noting the similarity between the EMPP view of the evolution of the real exchange rate and that implied by traditional relative PPP as given by condition (4). Superficially the two concepts appear to be very similar. Thus relative PPP as given by condition (4) implies that the change in the logarithm of the real exchange rate equals zero; here EMPPP tells the same kind of story, apart from a random error. Note, however, that the story underpinning condition (4) is radically different from that generating equation (10). In condition (4) it is goods arbitrage that gives relative PPP, whereas equation (16) exists through arbitrage on the capital account of the balance of payments. These differential theoretical underpinnings mean that the error term in equation (16) is far from an innocuous appendage. Thus, the existence of the error term indicates that a disturbance to the real exchange rate will have a permanent effect, whereas in condition (4) such disturbances are ruled out by definition. This difference may be seen more clearly in an example. In a world of short-run sticky prices, a monetary surprise, which pushes the nominal exchange rate away from the relative price relationship will, from equation (16), result in a permanent change in the real exchange rate. In the traditional Cassellian view the monetary disturbance, although not immediately forcing condition (2) or (4), would eventually lead to such relationships being restored; real exchange rates would be mean reverting.

<sup>14</sup> Where we have made use of the following expressions:

$$\begin{aligned} \Delta s_{t+k} &= E_t \Delta s_{t+k} + \epsilon_{t+k}, \quad \Delta p_{t+k} = E_t \Delta p_{t+k} + \varphi_{t+k}, \quad \Delta p_{t+k}^* = E_t \Delta p_{t+k}^* + \xi_{t+k} \quad \text{and} \\ \omega_{t+k} &= \epsilon_{t+k} - \varphi_{t+k} + \xi_{t+k}. \end{aligned}$$

<sup>15</sup> An alternative, less restrictive version of the EMPPP, would have interest rates being equalized up to a constant differential, in which case a constant would be added to the right-hand side of equation (16).

Also, the implications of equation (16) for the current account are rather odd since they suggest that the real exchange rate will only by chance be at a level consistent with a zero current balance. In equilibrium, therefore, the current account presumably does not matter. It would seem, therefore, that the assumption of perfect capital mobility is unreasonable, since it implies that a country can go on accumulating or decumulating net foreign assets without bound.

From a time-series perspective note that the derivation of equation (16) presupposes that real interest rates and the real exchange rates are integrated of different orders, or that the real exchange rate is  $I(1)$  and real interest rates are cointegrated (we shall return to this point again below).

### The Balance of Payments and the Specification of the Long-Run Exchange Rate Equation

Earlier in this section, we noted potential ways in which the balance of payments equilibrium condition may influence the specification of a long-run exchange rate model: productivity differences between countries or net asset accumulation can distort the exchange rate/relative price relationship. However, when comparing countries at a similar level of development, which nearly all of the existing research does (and which we do in this paper), this is unlikely to be important.<sup>16</sup> The foreign asset accumulation channel is unclear in the current context, although, again, for countries at a similar level of development it may not be that important. However, there is another way in which equation (8) may be useful in defining a long-run equilibrium exchange rate: this relates to the distinction between what may be referred to as a statistical and "true" long run (the latter concept has been referred to as a long long run by Breuer (1994)).

In many of the papers discussed in this survey, when a researcher refers to an estimated long-run exchange rate relationship, he or she is referring to the existence of a statistical equilibrium; that is, one that is consistent with the particular estimator used (for example, the Johansen maximum likelihood procedure). This may or may not conform to a "true" equilibrium position, which is one defined by economic theory. Indeed, the kind of long-run equilibrium that many researchers have estimated for the recent float does not actually conform to what most would understand by equilibrium (i.e., a true equilibrium). We suggest that the balance of

<sup>16</sup> Japan, however, is likely to be the exception—see Faruquee (1995).



payments equilibrium condition may be a useful way of understanding the empirical findings reported in this paper and reconciling them with the parameter values implied by the pure Cassellian notion of PPP. This is a topic to which we return in the concluding section.

### III. Modeling the Long-Run Nominal Exchange Rate

In this section we present an overview of the literature on modeling the long-run nominal exchange rate using cointegration methods. The section is in two parts. The first part contains a survey of the literature relating nominal bilateral exchange rates to relative price indices. The second contains an overview of estimated cointegrating relationships for the monetary model of the exchange rate.

#### Absolute Purchasing Power Parity and Cointegration

Much recent work on modeling equilibrium exchange rates has focused on testing equation (2'). Since the variables contained in (2') are likely to be nonstationary, such tests have concentrated on exploiting the cointegration methods proposed by Engle and Granger (1987). Since this and its associated methods are now well documented we will not give a detailed account here (see, for example, Banerjee and others (1993) for a comprehensive survey). A brief account is, however, beneficial not least from the point of view of introducing terminology, and we focus our discussion on equation (17), which is the regression equation analogue to (2'):

$$s_t = \beta + \alpha_0 p_t + \alpha_1 p_t^* + \varphi_t. \quad (17)$$

If the variables entering equation (17) are all first-order nonstationary,<sup>17</sup> then they are integrated of order 1,  $I(1)$ . If there is no long-run relationship between the exchange rate and relative prices, the residual series in equation (17) would also be nonstationary,  $I(1)$ .<sup>18</sup> If there is a long-run relationship between an exchange rate and relative prices, which we note from our previous discussion is what proponents of traditional PPP have

<sup>17</sup> This is not uncontroversial since there is some evidence to suggest that prices are  $I(2)$  processes (exchange rates being  $I(1)$ ); see, for example, MacDonald (1993).

<sup>18</sup> However, in such circumstances there may still be a short-run relationship and an appropriate way to capture this would be to specify the regression in first differences (which would amount to a test of relative PPP—see Frenkel (1981), Krugman (1978), and MacDonald (1988b) for such estimates).

in mind, then the appropriate way to capture it is to use cointegration methods. In the current context cointegration could be said to exist when, despite the variables entering equation (17) being individually non-stationary, there exists some linear combination that transforms the residuals to an  $I(0)$  series. There is now a plethora of different ways of testing for cointegration, each of which has been developed from the initial testing methods of Engle and Granger (1987). Cointegration-based tests of PPP have followed, in a chronological sense, the development of cointegration tests and, as we shall see, the more recent cointegration tests facilitate a more powerful test of PPP than the baseline Engle-Granger method.

In the context of the cointegration literature, the existence of long-run PPP amounts to the satisfaction of three conditions. First, and most important, the errors,  $\varphi_t$ , from an estimated version of equation (17) should be stationary; that is, they should be  $I(0)$ . If they are not, then there will be a tendency for the exchange rate and relative prices to drift apart without bound, even in the long run. Second, the  $\alpha_0$  and  $\alpha_1$  coefficients should enter equation (17) with an equal and opposite sign (the condition of symmetry) and, third, they should be equal to plus and minus unity (the condition of proportionality).<sup>19</sup>

The Engle-Granger cointegration method simply entails estimating equation (17) by ordinary least squares (OLS) and subjecting the residuals to a variety of diagnostic tests of which the most popular has proven to be the Augmented Dickey-Fuller test. This amounts to estimating an equation of the form

$$\Delta\varphi_t = v_1\varphi_{t-1} + v_i\sum_{i=2}^p\Delta\varphi_{t-i+1} + \epsilon_t. \quad (18)$$

If the null hypothesis of no cointegration is valid—the residuals are  $I(1)$ —then  $v_1$  should be insignificantly different from zero and this may be tested using a  $t$ -test, denoted  $\tau$ . Under the alternative hypothesis of stationarity  $v_1$  is expected to be significantly negative. As the distribution of  $\tau$  is nonstandard, Engle and Granger have tabulated the appropriate critical values (other sets of critical values are given by Engle and Yoo (1987) and MacKinnon (1991)). One advantage of the Engle-Granger approach, as highlighted by Stock (1987), is that if cointegration exists, then even if all the variables entering equation (17) are  $I(1)$ , the coefficient estimates approach their asymptotic values at a rate equal to  $T$  rather than the conventional  $T^{1/2}$ , where  $T$  denotes the number of obser-

<sup>19</sup>The distinction between symmetry and proportionality is, we feel, rather artificial, but is one that is often made in the literature.

valuations; that is, the estimates are super-consistent. This information may be helpful in allowing a researcher to gauge how far away from symmetry and proportionality her estimates are, although one problem with the Engle-Granger approach is that it does not allow one to draw any inferences on the basis of these values (thus, the fact that the variables entering equation (17) are all nonstationary means that standard statistical inference is not valid).

For the recent experience with flexible exchange rates, the Engle-Granger two-step cointegration method has been applied to aggregate price data by Enders (1988), Mark (1990), Patel (1990), and Taylor (1988). Because in their initial paper Engle and Granger only computed critical values for  $\tau$  for a regression equation with two variables, Enders, Mark, and Taylor constrain the coefficient on the relative price terms to be equal and opposite (that is they impose symmetry). A paper by Engle and Yoo (1987) tabulates critical values for  $\tau$  from a regression of up to five variables and these are used by Patel to estimate equation (17) in unconstrained fashion. Enders (1988) estimates equation (17) with relative wholesale price terms constructed for Canada, Germany, and Japan against the United States for the period January 1973 to November 1986 and is unable to reject the null of no cointegration in any instance. Taylor comes to the same conclusion using a similar data base. Mark investigates a number of OECD bilateral rates based on, respectively, the U.S. dollar, U.K. pound, and Japanese yen as the home currency for the period June 1973 to February 1988 (consumer prices) and finds only one instance (out of 13) when the null of no cointegration is rejected.<sup>20</sup> Patel uses a quarterly data base spanning the period 1974–86 for Canada, Germany, Japan, the Netherlands, and the United States (a variety of bilateral exchange rate combinations are considered for these countries) and reports that the null is rejected in only 4 instances out of a total of 15. In sum, we interpret this group of papers as suggesting that there is no long-run tendency for exchange rates and relative prices to settle down on an equilibrium track.<sup>21</sup> One disadvantage of these studies, as we indicated above, is that the use of the two-step methodology precludes an actual test of the proportionality and symmetry of the  $\alpha$ 's with respect to the exchange rate, although the estimated values are often far from 1 and -1.

In Table 1 we present some estimates of equations (17) and (18) using the data set discussed in the introduction. With the WPI as the price

<sup>20</sup> This one significant value is less than the number that would be expected to occur by chance for this group of exchange rates.

<sup>21</sup> Choudhry, McNown, and Wallace (1991), using the Engle-Granger method, find some evidence of cointegration for the Canadian dollar in the 1950s.

Table 1. *Engle-Granger Two-Step Cointegration Tests*

|                | CPI        |            |        | WPI        |            |        |
|----------------|------------|------------|--------|------------|------------|--------|
|                | $\alpha_0$ | $\alpha_1$ | ADF    | $\alpha_0$ | $\alpha_1$ | ADF    |
| Canada         | -0.0001    | 0.223      | -2.400 | 0.783      | -0.672     | 2.100  |
| France         | 3.157      | -3.662     | -2.260 | 1.709      | -1.192     | -2.350 |
| Germany        | 5.552      | -3.266     | -1.620 | 3.144      | -2.250     | -1.420 |
| Italy          | 1.916      | -2.892     | -2.050 | 0.668      | -0.365     | -1.470 |
| Japan          | 1.088      | -1.390     | -2.670 | 2.161      | -1.621     | -2.920 |
| Netherlands    | 1.681      | -1.275     | -2.150 | 2.111      | -1.593     | -1.310 |
| Sweden         | -0.318     | 0.921      | -1.950 | 0.929      | -0.809     | -1.180 |
| Switzerland    | 0.382      | -0.758     | -2.360 | 2.254      | -1.355     | -1.650 |
| United Kingdom | 0.614      | -0.579     | -2.350 | 0.517      | -0.478     | -2.060 |

Notes: The countries in the first column denote the home currency component of the nominal exchange rate used in the Engle-Granger two-step regression (in all cases the foreign currency is the U.S. dollar). The entries in the columns labeled  $\alpha_0$  and  $\alpha_1$  denote estimated coefficients and ADF denotes the augmented Dickey-Fuller statistic calculated from the residuals of the cointegration regression. The critical value for the latter is  $-2.98$  (source: MacKinnon (1991)). The labels CPI and WPI indicate the use of a consumer or wholesale price measure in the cointegrating regression.

measure we note that all of the  $\alpha_0$  and  $\alpha_1$  coefficients are correctly signed (positive and negative, respectively) while with the CPI seven of the nine currencies produce correctly signed values of these coefficients. However, most of the estimated coefficients are far from their hypothesized values of 1 and  $-1$ . Of most concern, though, is the fact that none of the estimated augmented Dickey-Fuller statistics are significant at the 5 percent level (indeed, none are significant at even the 10 percent level). These results confirm, albeit with a longer sample period, the research discussed in the last paragraph.

The single equation estimator of Engle and Granger, however, poses the researcher with very real practical problems. Thus, Banerjee and others (1986) have noted that the small sample properties based on an equation like (17) are poor. Additionally, if the regressors in (17) are endogenous (which, as we suggested earlier, they are highly likely to be in our application) and (or) the errors exhibit serial correlation (which again is very likely in the current application), then the asymptotic distribution of  $T(\hat{A} - A)$ , where  $T$  denotes the number of observations,  $A' = [\alpha_0 \ \alpha_1]$ , and  $\hat{A}$  is an estimate of  $A$ , will depend upon nuisance parameters. The Full Information Maximum Likelihood (FIML) method of Johansen (1988 and 1991) and Ahn and Reinsel (1988), which imposes unit roots on the variables, will produce asymptotically optimal estimates because they incorporate a parametric correction for serial correlation (which comes from the assumed underlying VAR structure—see below)

and, since it is a systems method, it can handle the endogeneity of the regressors (to the extent the implied price equations are plausible).

Since the method of Johansen is now well known we do not discuss it here. Rather, we simply note two test statistics that we and other researchers have used to test for the number of cointegrating vectors. In our application the likelihood ratio, or Trace, test statistic for the hypothesis that there are at most  $r$  distinct cointegrating vectors is

$$LRI = T \sum_{i=r+1}^N \ln(1 - \hat{\lambda}_i), \quad (19)$$

where  $\hat{\lambda}_{r+1}, \dots, \hat{\lambda}_N$  are the  $N-r$  smallest squared canonical correlations between  $X_{t-k}$  and  $\Delta X_t$  series (where  $X_t = (s_t, p_t, p_t^*)'$  and where all of the variables entering  $X$  are assumed  $I(1)$ ), corrected for the effect of the lagged differences of the  $X$  process (for details of how to extract the  $\lambda_i$ 's see Johansen (1988) and Johansen and Juselius (1990)). Additionally, the likelihood ratio statistic for testing at most  $r$  cointegrating vectors against the alternative of  $r+1$  cointegrating vectors—the maximum eigenvalue statistic—is given by

$$LR2 = T \ln(1 - \lambda_{r+1}). \quad (20)$$

Johansen (1988) shows that equations (19) and (20) have a nonstandard distribution under the null hypothesis. He does, however, provide approximate critical values for the statistic, generated by Monte Carlo methods.

A number of researchers have argued that the failure to find a cointegrating relationship between relative prices and an exchange rate may be due to the econometric method used, rather than the absence of a long-run relationship. For example, Cheung and Lai (1993), Kugler and Lenz (1993), MacDonald (1993), and MacDonald and Marsh (1994a) all advocate using the Johansen cointegration method to test for the number of cointegrating vectors among relative prices and exchange rates for bilateral U.S. dollar exchange rates (MacDonald (1993); MacDonald and Marsh (1994a); and Cheung and Lai (1993)) and German mark bilateral dollar rates (Kugler and Lenz (1993); MacDonald (1993); and MacDonald and Marsh (1994a)). A considerable amount of evidence in these papers supports the contention that there is indeed a long-run PPP relationship for a variety of currencies in the sense that most bilateral currency/price combinations exhibit cointegration. However, often the restrictions of symmetry and proportionality are rejected in these studies (especially when U.S. dollar bilateral rates are used).

In Table 2 we present estimates of the PPP relationship for our data set using the Johansen method. The table should be read in the following

Table 2. *Johansen Multivariate Cointegration Tests*

|         | Trace |       | $\lambda$ Max |      | $\alpha_0$ | $\alpha_1$ | $\gamma$ | LR3   | LR4    |       |        |
|---------|-------|-------|---------------|------|------------|------------|----------|-------|--------|-------|--------|
| Canada  |       |       |               |      |            |            |          |       |        |       |        |
| CPI     | 5.43  | 13.65 | 35.54*        | 2.53 | 8.22       | 21.88*     | -10.88   | -0.12 | -0.02  | 11.9* | 12.2*  |
| WPI     | 2.53  | 9.76  | 39.72*        | 2.53 | 7.23       | 29.96*     | -1.12    | 0.01  | -0.01  | 15.0* | 25.0*  |
| France  |       |       |               |      |            |            |          |       |        |       |        |
| CPI     | 4.04  | 15.02 | 35.45*        | 4.04 | 10.98      | 20.43      | -2.95    | 0.03  | -0.07  | 0.1   | 0.3    |
| WPI     | ...   | ...   | ...           | ...  | ...        | ...        | ...      | ...   | ...    | —     | —      |
| Germany |       |       |               |      |            |            |          |       |        |       |        |
| CPI     | 0.04  | 6.59  | 19.62         | 0.04 | 6.56       | 13.03      | -0.37    | 0.01  | -0.02  | —     | —      |
| WPI     | 0.00  | 9.26  | 24.16         | 0.00 | 9.25       | 14.90      | -84.6    | 0.46  | -0.00  | —     | —      |
| Italy   |       |       |               |      |            |            |          |       |        |       |        |
| CPI     | 6.17  | 14.91 | 35.79*        | 6.17 | 8.74       | 20.88      | -5.8     | 0.09  | -0.01  | 10.6  | 11.1*  |
| WPI     | 0.82  | 6.25  | 25.06         | 0.82 | 5.43       | 18.81      | 7.11     | -0.10 | -0.01  | —     | —      |
| Japan   |       |       |               |      |            |            |          |       |        |       |        |
| CPI     | 4.64  | 18.29 | 45.98*        | 4.64 | 13.65      | 27.69*     | -22.9    | 0.11  | 0.01   | 0.02  | 6.18   |
| WPI     | 1.71  | 12.18 | 31.87*        | 1.71 | 10.46      | 19.68      | -2.42    | 1.78  | -0.04* | 4.28* | 15.29* |

## Netherlands

|     |      |       |        |      |       |        |       |       |       |       |        |
|-----|------|-------|--------|------|-------|--------|-------|-------|-------|-------|--------|
| CPI | 4.21 | 17.15 | 36.92* | 4.21 | 12.73 | 19.77  | 6.60  | -0.00 | -0.01 | 0.01  | 5.51   |
| WPI | 1.13 | 11.49 | 36.30* | 1.13 | 10.36 | 24.90* | 20.46 | -0.10 | -0.01 | 9.09* | 15.97* |

## Sweden

|     |      |       |       |      |       |       |      |       |       |   |   |
|-----|------|-------|-------|------|-------|-------|------|-------|-------|---|---|
| CPI | 0.07 | 17.53 | 23.37 | 1.07 | 4.76  | 17.53 | 2.84 | -0.04 | 0.07  | — | — |
| WPI | 2.34 | 12.52 | 26.52 | 2.34 | 10.19 | 13.99 | 8.49 | -0.13 | -0.01 | — | — |

## Switzerland

|     |      |       |        |      |       |        |      |       |       |       |        |
|-----|------|-------|--------|------|-------|--------|------|-------|-------|-------|--------|
| CPI | 0.16 | 7.60  | 27.03  | 0.16 | 7.43  | 19.43  | 2.28 | -0.00 | -0.05 | 6.23* | 11.27* |
| WPI | 0.92 | 11.38 | 41.43* | 0.92 | 10.46 | 30.05* | 4.38 | 0.00  | -0.01 | 9.67* | 19.41* |

## United Kingdom

|     |       |       |        |       |       |        |       |      |       |      |       |
|-----|-------|-------|--------|-------|-------|--------|-------|------|-------|------|-------|
| CPI | 10.08 | 24.05 | 43.79* | 10.08 | 13.98 | 19.74  | -0.59 | 0.01 | -0.06 | 0.34 | 6.96* |
| WPI | 3.35  | 16.28 | 40.91* | 3.35  | 12.93 | 24.63* | 0.27  | 0.01 | -0.03 | 0.01 | 3.17  |

Notes: The first column indicates the country used in the Johansen test and the relevant price series. Entries in the columns directly below TRACE and  $\lambda$ Max are the estimates of the Trace (19) and  $\lambda$ Max (20) statistics discussed in the text. The estimates of the normalized (on the exchange rate) cointegration coefficients are contained in the two columns headed by  $\alpha$ , and the entries in the  $\gamma$  column are the estimated adjustment coefficients from the exchange rate equation. LR3 and LR4 are, respectively, likelihood ratio tests for symmetry and proportionality. An asterisk denotes significance at the 5 percent level.

way. The columns under the heading "Trace" contain our estimates of equation (19), while the entries in the columns under " $\lambda$ Max" contain our estimates of equation (20). The estimates of the normalized cointegration vector are contained in the two columns under  $\alpha$ , the entries in the  $\gamma$  column are the estimated adjustment coefficients from the exchange rate equation, and the entries in the *LR3* and *LR4* columns are likelihood ratio test statistics for testing, respectively, proportionality and symmetry. An asterisk denotes that a statistic is significant at the 5 percent significance level. Note that on the basis of the Trace and  $\lambda$ Max statistics there is evidence of at least one cointegrating vector for each currency apart from Sweden and Germany.<sup>22</sup> Note, further, that although many of the estimated coefficients (in the  $\alpha$  columns) are correctly signed, there are a number that are wrongly signed, and also many coefficient values are far from their numerical values of unity, in absolute terms. It is not surprising, therefore, that the proportionality and symmetry restrictions are convincingly rejected for most currency-price combinations.

One interesting feature of the results reported in Table 2 is the speed with which the exchange rate adjusts to its equilibrium value. Apart from two of the estimated  $\gamma$ 's, all of the coefficients are negative (and only one of the positive values is statistically significant). The average of the adjustment speeds across currencies (for those currencies that produce a negative adjustment coefficient,  $\gamma$ ) is approximately  $-0.02$ , which indicates that, on average, 2 percent of a deviation from the error correction mechanism (ECM) is corrected within a month. On the basis of this number the half-life of a deviation from the ECM is 36 months (that is,  $(1-0.02)^{36}$ ). Compared with some of the adjustment speeds discussed below, this represents a relatively rapid adjustment back to equilibrium. However, it is important to note that the adjustment here does not represent an adjustment to strict PPP (that is, where proportionality holds), but rather to a mongrelized version of PPP in which the coefficients on relative prices are not at their hypothesized values.

Why are the  $\alpha$  coefficients in an estimated version of equation (17) noted here, and also in the work of the other researchers discussed above, often so far away from their hypothesized values? After all, cointegration methods should provide super-consistent results, free from problems of omitted variable bias and endogeneity. One explanation that has been offered by Taylor (1988), Patel (1990), Cheung and Lai (1993), and MacDonald (1993) is that the rejection of the symmetry and proportion-

<sup>22</sup>The results for Germany differ from those reported in MacDonald (1993). The difference seems to stem from the longer sample used in this paper, thereby indicating a potential instability for this currency.



ality hypotheses may stem from the existence of trending measurement error and trends in the relative prices of traded to nontraded goods. Indeed, Patel (1990) asserts that all one can say about the  $\alpha_0$  and  $\alpha_1$  coefficients in an estimated version of equation (17) is that they should be positive and negative, respectively. It would seem implausible, however, to believe that the large deviations of the coefficients reported in Table 2 from their priors is due to such trends. Intriguingly, though, MacDonald in his cointegration study computed the cross rates of the European currencies represented in his sample and was unable to reject proportionality for three out of the six combinations. Given that the existence of the European Union (EU) presumably implies greater similarity in the construction and constituents of price indices for EU participants, this may be suggestive that measurement error and/or traded/nontraded biases at least partly explains the results for U.S. bilateral exchange rates. An alternative explanation lies in the econometrics of cointegration.

As we have noted, the small sample properties of the coefficient estimates, particularly those based on the Engle-Granger method, are likely to suffer from biases arising from simultaneity between prices and exchange rates and also serially correlated disturbances. Although the Johansen method, in principle, addresses these issues, in practice the specification of the equations for the price levels in the home and foreign country may not be adequate and this may, in turn, introduce misspecification into the exchange rate equation (since the Johansen method is a full information maximum likelihood procedure). An alternative method for handling these problems is the fully modified single equation estimator of Phillips and Hansen (1990). This uses a nonparametric correction to produce estimates of the coefficients and standard errors in equation (17) that are fully modified in the sense that they are robust to serial correlation of the disturbances and to potential endogeneity of the regressors. Hansen (1992) has developed this estimator by deriving an algorithm that allows one to interpret rejections of the null hypothesis of cointegration in terms of coefficient instability. In the present application the idea may be seen in the following way.

As we have noted, if  $s_t$ ,  $p_t$ , and  $p_t^*$  are cointegrated, the error term,  $\varphi_t$ , in equation (17) should be  $I(0)$ . If, however,  $s_t$ ,  $p_t$ , and  $p_t^*$  are not cointegrated then  $\varphi_t$  is  $I(1)$  and we may think of it as containing a random walk component,  $D_t$ , and a stationary term,  $v_t$  (i.e.,  $\varphi_t = D_t + v_t$ ). Under these conditions we may rewrite equation (17) as

$$s_t = \beta_t + \alpha_0 p_t + \alpha_1 p_t^* + v_t, \quad (17')$$

where  $\beta_t = \beta + D_t$ . Hence the alternative hypothesis is equivalent to the intercept term in equation (17) following a random walk.

MacDonald and Moore (1994) implement the Phillips-Hansen and Hansen method for three groups of bilateral currencies, based on the German mark, Japanese yen, and U.S. dollar (consumer price indices are used in the comparisons), using data for the recent float. The other countries involved in these bilateral relationships are Canada, France, Italy, the Netherlands, Sweden, Switzerland, and the United Kingdom. Overall, their results may be summarized by saying they indicate a remarkable degree of stability for all three bilateral groupings, in the sense that  $\beta$  does not follow a random walk. Interestingly, it is only the Japanese yen-U.S. dollar relationship that exhibits any evidence of instability (a finding that MacDonald and Moore attribute to persistent long-term capital flows from Japan to the United States over the period and the continuing productivity bias in favor of Japanese goods).<sup>23</sup> Additionally, in 16 out of 24 instances the symmetry restriction cannot be rejected and in 7 out of 24 instances the proportionality hypothesis cannot be rejected either. Overall this is a better strike record than that reported in Table 2, and may suggest that the econometric method used in estimating the cointegrating relationship is important in defining the  $\alpha$  coefficients.

The empirical evidence relating to equation (17) may be summarized in two ways. First, there is now mounting evidence to suggest that the residual in an estimated version of equation (17) is a mean-reverting process; that is, it is stationary and the deviations from the estimated ECM seem to be relatively short lived. Second, the relationship between exchange rates and relative prices rarely obeys the symmetry and degree one homogeneity restrictions, which is suggestive of real factors (such as net asset accumulation, perhaps) requiring real exchange rate adjustments for the particular sample periods considered by the above researchers. We return to these issues in the concluding section.

### The Nominal Exchange Rate and Relative Excess Money Supplies

A long-run model of the exchange rate that is closely related to PPP is the flex-price monetary model. Useful surveys of this approach are to be found in Frenkel and Mussa (1985), MacDonald (1988a), (1990),

<sup>23</sup> This finding is consonant with Faruquee's (1995) results for the real effective value of the Japanese yen over a period encompassing the floating period and part of the Bretton Woods period. It is also consistent with the adjustment coefficient for the Japanese PPP relationship, based on the CPI, reported in Table 2.

MacDonald and Taylor (1992), and Frankel and Rose (1994)). This model essentially appends a theory of the determination of the price level to absolute PPP. In particular, assume that the domestic and foreign demand for money functions may be written using standard Cagan log-linear specifications:

$$m^D - p = \beta_0 y - \beta_1 i, \quad (21)$$

$$m^{D*} - p^* = \beta_0 y^* - \beta_1 i^*, \quad (21')$$

where, of terms not previously defined,  $m$  is the logarithm of the demand for money,  $y$  is the logarithm of income,  $i$  is the nominal interest rate, and  $\beta_0$  and  $\beta_1$  are, respectively, the income elasticity and the interest semi-elasticity of the demand for money (we have assumed these to be identical across countries). The money market equilibrium conditions for the home and foreign country are given by

$$m^D = m, \quad (22)$$

$$m^{D*} = m^*, \quad (22')$$

where  $m$  denotes the supply of money. On using equation (22) in (21) and solving for the relative price level, we obtain the long-run relative price relationship

$$p - p^* = m - m^* - \beta_0(y - y^*) + \beta_1(i - i^*), \quad (23)$$

which posits that the relative price of home to foreign goods is determined by the excess of money supply over money demand. On substituting this expression into equation (16) we obtain

$$s = m - m^* - \beta_0(y - y^*) + \beta_1(i - i^*), \quad (24)$$

which is the (continuous) solution for the flex-price monetary model (Frenkel (1976) and Hodrick (1978)) and the long-run solution for the sticky price model (see Dornbusch (1976), Frankel (1979), and Buiter and Miller (1981)).

A number of researchers have tested equation (24), or variants thereof, using the Engle-Granger two-step procedure. For example, Boothe and Glassman (1987) test for cointegration of the U.S. dollar/deutsche mark exchange rate and only the relative money supply and are unable to reject the null hypothesis of no cointegration. However, it is not entirely clear that Boothe and Glassman exploit a potentially valid cointegrating set since the appropriate long-run monetary model emphasizes relative excess money supplies; that is, relative money supplies adjusted for, at

least, relative incomes and perhaps also interest rates.<sup>24</sup> Meese (1986) tests the monetary model for U.S. bilateral dollar rates of the deutsche mark, the pound sterling, and the yen, but he is unable to unearth a valid cointegrating set.<sup>25</sup> Kearney and MacDonald (1990) test for cointegration between the Australian dollar/U.S. dollar exchange rate and relative money, income, and interest rates and are unable to reject the null of no cointegration. Finally, Baillie and Selover (1987) test whether a version of the sticky price variant of the monetary model is able to produce a valid cointegrating set, for the bilateral U.S. dollar rates of the Canadian dollar, French franc, German mark, Japanese yen, and U.K. pound. In common with the other Engle-Granger-based studies, these authors are also unable to reject the null of no cointegration. The combined impression one obtains from the above-noted results would therefore seem to suggest that the monetary model does not even have empirical support as a long-run relationship.

However, paralleling the recent cointegration literature on PPP, MacDonald and Taylor (1991) have criticized the use of the two-step procedure to test the monetary model and have, instead, advocated the Johansen (1988) maximum likelihood method. Interestingly, in using this approach to test the monetary model for three currencies (dollar-mark, dollar-sterling, and dollar-yen), MacDonald and Taylor demonstrate that there is very strong support for the monetary model as a long-run relationship (indeed for the dollar-mark exchange rate they show that all of the restrictions implied by the monetary model are accepted by the data).<sup>26</sup> The average adjustment speed across the three currencies is  $-0.06$ , which is faster than the average adjustment coefficient from the PPP equations. Moosa (1994) also uses the Johansen method to test the validity of the monetary model for the pound sterling, mark, and yen (against the U.S. dollar) for the period January 1975 to December 1986. Moosa's estimated version of the monetary model differs from that of MacDonald and Taylor in that he distinguishes between traded and nontraded goods in his specification of equation (23). This extension to

<sup>24</sup> In the forward-looking, or rational expectations, version of equation (24), the exchange rate is the present discounted value of expected future relative money supplies and income levels and the appropriate cointegrating relationship does not include interest rates. The static version of the monetary model, given by equation (24), suggests that interest rates should be included in the cointegrating set.

<sup>25</sup> Meese tests the forward monetary model and has, in his cointegrating set, the exchange rate and relative money supplies and income levels.

<sup>26</sup> In particular, they find evidence of degree one homogeneity between the exchange rate and relative money supplies and coefficients on relative income and interest rate terms that are correctly signed and have plausible magnitudes.

the model also produces strong evidence of cointegration (although he finds that the monetary restrictions are rejected).

#### IV. Modeling Long-Run Real Exchange Rates: Efficient Markets and the Random Walk Real Exchange Rate Model

An alternative to testing for cointegration between a nominal exchange rate and relative prices is to actually impose the symmetry and homogeneity restrictions and test if a real exchange rate contains a unit root. The null hypothesis therefore is given by equation (16), which we repeat here with a drift term

$$\Delta q_t = a + \omega_t, \quad (16')$$

where  $\Delta$  is the first difference operator,  $a$  is a drift term, which captures, perhaps, the failure of real interest rates to be equalized across countries, and  $\omega_t$  is a stationary process. An alternative hypothesis to equation (16') is that the real exchange rate exhibits temporary deviations around a trend, that is, it is trend stationary:

$$q_t = \gamma_0 + \gamma_1 t + \epsilon_t, \quad (25)$$

where  $t$  denotes the time trend. The null hypothesis may be thought of as the efficient markets null, while the alternative hypothesis may be interpreted as traditional absolute PPP.

The standard test of the above null hypothesis against the trend stationary alternative may be understood using the following autoregressive moving average (ARMA) specification for the real exchange rate (as may some of the other tests considered in this section):

$$\phi(B)q_t = a + \theta(B)\epsilon_t, \quad (26)$$

where  $B$  denotes the lag operator and  $a = \gamma_0 + \gamma_1 t$ . The following sets of tests are all dependent in some form or other on equation (26).

The easiest way to motivate a test for a unit root in  $q_t$  is to assume that the real exchange rate has a purely autoregressive representation, which will be the case if the moving average polynomial,  $\theta(B)$ , in equation (26) is invertible. Given this assumption we may reparameterize equation (26) as

$$\Delta q_t = \gamma_0 + \gamma_1 t + (\beta_0 - 1)q_{t-1} + \sum_{j=1}^{n-1} \beta_j \Delta q_{t-j} + v_t, \quad (26')$$

where

$$\beta_i = \sum_{j=1}^n \phi_j, \quad i = 1, \dots, n.$$

Since  $\phi(B)$  will contain a unit root if  $\sum_i \phi_i = 1$ , the presence of a unit root is formally equivalent to a test of whether  $\beta_0 = 1$  or  $(\beta_0 - 1) = 0$ . This hypothesis may be tested using a standard  $t$ -test, although as Dickey and Fuller (1979) and many others note, this will have a nonstandard distribution and therefore one has to use the percentiles tabulated by Fuller (1976). One may test for two unit roots in the real exchange rate by estimating equation (26') with all of the real exchange rate terms first differenced again. The inclusion of the correct specification of deterministic variables in equation (26') is crucial to the power of the test (see Banerjee and others (1993)).

Much as in the cointegration literature on testing for absolute PPP, papers that test for a unit root in real exchange rates can be placed into two groups; those in the first, chronologically earlier, group yield evidence that is favorable to the hypothesis, while those in the second, more recent, group are unfavorable. The first group essentially uses standard (classical) unit root test statistics, as given by equation (26') to test the null hypothesis that a variety of bilateral and effective real exchange rates contain a unit root (see, *inter alia*, Roll (1979), Frenkel (1981), Darby (1980), Adler and Lehmann (1983), MacDonald (1985a and 1985b), Meese and Rogoff (1988), Enders (1988), Mark (1990), and Edison and Pauls (1993)).<sup>27</sup> The vast majority of such tests, which all use monthly data, are unable to reject the null.

In Table 3 we illustrate the findings of this group of papers by presenting our tests for one and two unit roots in our real exchange rate data.<sup>28</sup> We constructed two real exchange rates for each country: one based on the CPI, the other based on the WPI. Each real exchange rate is expressed in natural logarithms. The results, regardless of the deterministic specification used, strongly indicate the presence of a single stochastic unit root. These results are therefore consistent with the results of other researchers noted in the previous paragraph. Although we cannot reject the hypothesis that  $\beta_0$  is unity, we also report in Table 3 the estimated value of  $(1 - \beta_0)$  from the levels autoregression (without a time trend). The average value of this number is approximately  $-0.03$ , which is very close to the average adjustment estimate using the Johansen method (although of course the estimates are not strictly comparable).

<sup>27</sup> The papers of Roll (1979) and Adler and Lehmann (1983) both use an autoregression of the differenced real exchange rate to test the hypothesis that the sum of the coefficients on the autoregressive terms are jointly equal to zero.

<sup>28</sup> See Adams and Chadha (1991) for a discussion of the unit root properties of four U.S. dollar bilateral exchange rates, and some fundamental determinants of exchange rates over the period 1974, quarter 1, to 1989, quarter 4.

Table 3. *Tests for a Unit Root in the Real Exchange Rate Series*

|                |     | L     |        |                 | $\Delta L$ |        |
|----------------|-----|-------|--------|-----------------|------------|--------|
|                |     | $t$   | $\tau$ | $(1 - \beta_0)$ | $t$        | $\tau$ |
| Canada         | CPI | -2.18 | -2.19  | -0.02           | -3.54      | -3.76  |
|                | WPI | -2.14 | -2.11  | -0.04           | -3.93      | -4.08  |
| France         | CPI | -1.92 | -1.93  | -0.03           | -3.21      | -3.23  |
|                | WPI | ...   | ...    | ...             | ...        | ...    |
| Germany        | CPI | -2.00 | -1.89  | -0.03           | -3.15      | -3.23  |
|                | WPI | -1.76 | -1.80  | -0.03           | -3.29      | -3.43  |
| Italy          | CPI | -1.72 | -2.05  | -0.02           | -3.17      | -3.29  |
|                | WPI | -1.23 | -1.69  | -0.02           | -3.22      | -3.46  |
| Japan          | CPI | -1.92 | -2.82  | -0.02           | -3.51      | -3.51  |
|                | WPI | -2.87 | -2.01  | -0.03           | -3.82      | -3.88  |
| Netherlands    | CPI | -1.98 | -1.98  | -0.03           | -3.17      | -3.18  |
|                | WPI | -1.66 | -1.58  | -0.02           | -3.21      | -3.33  |
| Sweden         | CPI | -1.59 | -1.61  | -0.01           | -3.03      | -3.19  |
|                | WPI | -1.28 | -1.23  | -0.02           | -3.35      | -3.54  |
| Switzerland    | CPI | -2.25 | -2.31  | -0.04           | -3.61      | -3.59  |
|                | WPI | -2.07 | -2.07  | -0.04           | -3.84      | -3.87  |
| United Kingdom | CPI | -2.29 | -2.32  | -0.03           | -3.33      | -3.35  |
|                | WPI | -1.36 | -2.13  | -0.02           | -3.28      | -3.33  |

Notes: The exchange rates are real bilateral U.S. dollar rates of the countries listed in the first column and the price levels used to construct a real rate—CPI or WPI—are defined in the second column. The numbers in the columns headed  $t$  and  $\tau$  are, respectively, the estimated  $t$ -ratios from equation (26') when a constant ( $t$ ) and constant plus time trend ( $\tau$ ) are included as the deterministic variables. The two columns under L denote a test for a unit root in the level of the real exchange rate, while the two columns under  $\Delta L$  denote a test for a unit root in the first difference of the real rate. The 5 percent critical values for  $t$  and  $\tau$  are, respectively, -2.79 and -3.09 (source: Dickey and Fuller (1979)).

Indeed, it is now widely accepted that the kind of unit root test used in Table 3 has relatively low power in rejecting the null of a unit root, this being especially so when the numerical value of  $\beta_0$  is close to unity (see Campbell and Perron (1991)). How then should these results be interpreted? Do they really indicate that EMPPP is indeed a valid description of the evolution of the real exchange rate? In fact, much as in the cointegration literature, the above-noted unit root tests may be regarded as rather primitive and analogous to the original Engle-Granger two-step tests in their power at detecting the "true" underlying relationship. We

now consider a number of ways in which the above unit root tests are not the last word on the stochastic properties of the real exchange rate.

### The Power of Unit Root Tests and the Span of the Data

It has long been recognized that relatively high frequency data, such as monthly data, have a low signal-to-noise ratio compared with annual data. If a researcher were to conduct real exchange rate unit root tests for the recent float with annual data, there would be approximately 23 available observations. In order to increase the power of the test the researcher may be tempted to switch to a monthly data base, thereby increasing the number of observations to 276. However, in switching to the higher frequency data all that would likely have been achieved is the introduction of extra noise, without improving the precision of the point estimate (indeed, Frankel (1986) has demonstrated that this is the case in the context of testing for a unit root in real exchange rates). A better alternative would be to stick with annual data and expand either the time-series or cross-sectional dimensions. The improvements in power that extending the span of the data set may bring can be demonstrated by asking the question: how much annual data would be required to reject the null of a unit root in the real exchange rate?

Following Frankel (1986), the asymptotic standard error of  $\beta_0$  may be defined as being approximately equal to  $[(1 - \beta_0^2)/X]^{1/2}$ , where  $X$  represents the total number of observations. If, with 23 years of annual data, the estimated value of  $\beta_0$  is 0.85 per annum then the standard error with 23 years of annual data would be approximately 0.11, insufficient to reject the null hypothesis (i.e., this gives an implied  $t$ -ratio for the hypothesis that  $\beta_0 = 1$  of 1.34, which, even using a conventional  $t$  distribution, is not significant). Say, as an alternative, the researcher had access to 100 annual observations (where these have been obtained either from extending the time-series or cross-sectional dimensions of the data—these alternatives are considered in more detail below). With the value of  $\beta_0$  remaining at 0.85 the standard error would now be approximately 0.05, a value sufficient to reject the null hypothesis of a unit root. A value for  $\beta_0$  of 0.85 means that a shock that disturbs the PPP relationship has a half-life of approximately four years. As we have noted, the span of the data may in principle be extended in two different directions: by increasing either the cross-sectional or the time-series dimension.

The first set of tests of the importance of the span of the data on the mean-reverting properties of real exchange rates was conducted by Edison (1987) and by Frankel (1986), (1988). Edison employs annual data



over the period 1890 to 1978, for the U.S. dollar–pound sterling exchange rate, to test PPP using a dynamic error correction model. She finds, *inter alia*, that the coefficient on the change in relative prices is insignificantly different from unity and the solution for the long-run exchange rate yields absolute PPP. The estimated coefficient on the error correction term suggests that 9 percent of any deviation from PPP is extinguished within a year.

Frankel demonstrated that when he moved from a monthly post-Bretton Woods data base to a long-run annual data base, for the U.K. pound–U.S. dollar (period 1869–87), the estimated  $\rho$  coefficient from equation (10) changed from being insignificantly different from unity to significantly below unity. Interestingly, though, the real exchange rate series, even over this long time span, contained considerable persistence in the sense that only 16 percent of a deviation is extinguished per year. Grilli and Kaminsky (1991) also utilize annual data for the U.S. dollar–U.K. pound real exchange rate, over the period 1885–86, and test for a unit root using Phillips-Perron adjusted Dickey-Fuller statistics.<sup>29</sup> They find that the null of a random walk is rejected for the full sample period but not for a variety of subsamples.

Kim (1990) employs annual data for a wider range of currencies than the studies of Frankel and of Grilli and Kaminsky. In particular, he examines the U.S. bilateral real rates of the Canadian dollar, French franc, Italian lira, Japanese yen, and U.K. pound for the period 1900–87 (using CPI data) and 1914–87 (using WPI data). He demonstrates that the null hypothesis of a random walk may be rejected in all cases apart from the CPI-based real Canadian dollar, yen, and pound. An interesting finding in the light of the panel unit root tests discussed below is that the null of a unit root can only be rejected for the yen-dollar rate when a time trend is included in the regression.<sup>30</sup> The results of Frankel, Grilli and Kaminsky, and Kim would seem to support the view that having as long a time span as possible is important in discriminating between unit root and near unit root behavior.<sup>31</sup> Lothian and Taylor (1995) present some unit root tests for the real exchange rate of the U.S. dollar–pound sterling

<sup>29</sup>They also use variance ratio tests that are supportive of the Phillips-Perron tests—see our discussion below.

<sup>30</sup>Kim also conducts cointegration tests using this larger span of data and is able to reject the null of noncointegration (using the Engle-Granger two-step method) in six out of ten currency/price combinations. Furthermore, he obtains estimated coefficients on the price ratios that are numerically very close to unity for all of the cointegrating sets.

<sup>31</sup>Some other papers that also use long runs of data to test for unit roots are considered in the next section.

rate (annual data, 1791–1990) and French franc–pound sterling (annual data 1803–1990). Evidence of significant mean reversion is reported.

One potential hazard in using long runs of the time-series data is that the basket of goods used to construct the price measure may have changed beyond recognition between the start and end points of the sample. This kind of problem may be thought of as the temporal analogue to a point in time comparison, in which the construction of price series differs across countries (captured by the  $\pi$  term in equation (2')). Since the above tests will of course suffer from the latter problem as well, their usefulness must be qualified somewhat. A set of tests that also suffer from the spatial problem, but not the temporal (at least not to the same extent), are those that exploit the cross-sectional variability in prices and exchange rates. We now turn to this line of research.

MacDonald (1988b) uses annual average data on relative prices and exchange rates for the G-5 countries over the recent floating period and a pooled estimator to obtain the requisite degrees of freedom. (Hakkio (1984) suggested using cross-sectional data to increase the power of PPP tests based on a monthly data set for the recent float).<sup>32</sup> The results are interesting in that the symmetry/proportionality restrictions cannot be rejected. A representative result is reported here, with *t*-ratios in parentheses:

$$\Delta s_t = -1.199 - 1.040\Delta(p - p^*)_t, \quad (27)$$

(8.37) (3.58)

where the exchange rates are defined as the foreign currency price of a unit of home currency (in particular, these are U.S. dollar–domestic currency rates) and therefore the coefficient on relative prices is expected to be  $-1$  rather than  $+1$ . In fact, the hypothesis that the slope coefficient is  $-1$  cannot be rejected (the appropriate *t*-ratio is 0.13).<sup>33</sup> Flood and Taylor (1994) have also used a time-averaged/pooled approach in analyzing relative PPP for the recent float, and demonstrate that the type of time averaging employed is crucial in obtaining the proportionality result. In particular, using annual data over the period 1973–92 for 21 industrial countries, they demonstrate that as they move from one-year annual average data to ten-year averages the slope coefficient in an analogous expression to (27) moves from  $-0.109$  to  $0.967$ . The difference between Flood and Taylor's one-year average result and those reported by MacDonald perhaps reflects the different countries in the panel.

<sup>32</sup> As the PPP relationship is parsimonious regarding degrees of freedom, it may be argued that even for the recent float there are sufficient degrees of freedom to estimate it using annual data.

<sup>33</sup> The pooled timewise autoregressive/cross section heteroscedastic-consistent estimator of Kmenta (1986) was used to estimate equation (27).

The idea of increasing the data span of an annual data set by using pooled cross-sectional time-series data has been applied to the construction of unit root tests by Quah (1994) and Levin and Lin (1994). The latter authors, for example, demonstrate that implementing a unit root test on a pooled cross-section data set, rather than performing separate unit root tests for each individual series, can provide "dramatic improvements in statistical power." The Levin-Lin test is designed to evaluate the null hypothesis that each individual series is  $I(1)$ , versus the alternative hypothesis that all the series considered as a panel are stationary. The Levin-Lin test may be viewed as especially attractive since it facilitates the incorporation of a wide variety of individual-specific, or heterogeneous, effects under the null. Their testing method produces a single  $t$ -ratio for the panel and this statistic is shown to have a standard normal distribution. MacDonald (1995b) applies the Levin and Lin method to two panel data sets consisting of real exchange rates, defined using both WPI and CPI price measures, for the recent float.<sup>34</sup> The results turn out to be similar to those using long runs of annual data: the null hypothesis that each real exchange rate contains a unit root is rejected in favor of the alternative hypothesis that real exchange rates are stationary. Frankel and Rose (1995) test relative PPP and the unit root properties of real exchange rate in the context of an annual panel data set of 150 countries, for the period 1948–92. For the sub-sample period that corresponds to the recent float, they find that their estimated value of  $1 - \beta_{ii}$ , from the estimated panel version of equation (26'), is a statistically significant  $-0.15$ , implying a half-life of four years (the different estimation method used by MacDonald did not facilitate the extraction of a numerical estimate of panel mean reversion). This implied half-life is very similar to half-lives calculated from annual data sets spanning long time periods (see Frankel's results as reported above). Our discussion in this section has raised the issue of the power of unit root tests. We now focus our attention on alternative methods of testing for a unit root that may be more powerful in their ability to reject the null of a unit root.

### Alternative Unit Root Tests

In the last subsection our explanation for the finding of a unit root in real exchange rates lay in the type of data set used by researchers. An alternative, although not mutually exclusive explanation, is to be found

<sup>34</sup> More specifically, the data sets are for OECD currencies and consist of 17 real rates based on the WPI and 23 real rates based on the CPI, over the period 1973 to 1992 (annual data).

in the type of unit root test employed by the researchers noted above. As is now increasingly well-known, one disadvantage of unit root tests based on equation (26') or variants thereof is that they have relatively low power to test alternatives of near-stationary behavior (see Cochrane (1988) and Campbell and Perron (1991)). One way of thinking about this is to say that it may take a long time for real rates to exhibit mean-reverting behavior and such behavior will certainly not be picked up by the lag lengths conventionally used in an estimated version of equation (26').<sup>35</sup> A better way of picking up long autocorrelations may be to use the variance ratio test, recently introduced into the economics literature by Cochrane. This test uses the insight that if a series does indeed follow a random walk (the null hypothesis) then the variance of the  $k$ th difference of the series should equal  $k$  times the first difference. That is, if equation (16) truly is the time-series representation of the real exchange rate, then

$$\text{Var}(q_t - q_{t-k}) = k \text{Var}(q_t - q_{t-1}), \quad (28)$$

or

$$V_k = \frac{1}{k} \frac{\text{Var}(q_t - q_{t-k})}{\text{Var}(q_t - q_{t-1})} = 1, \quad (28')$$

where  $V_k$  denotes the variance ratio, based on lag  $k$ .<sup>36</sup> Lo and MacKinlay (1988) have demonstrated that the variance ratio is asymptotically equal to 1 plus a weighted average of the first  $k-1$  autocorrelation coefficients of  $q_t - q_{t-1}$ . If the average of these autocorrelations is zero,  $V_k$  will be unity. If, however, there is a preponderance of negative autocorrelations this will produce a value of  $V_k$  less than one and the real rate will display mean reversion. Conversely, if positive autocorrelations predominate this will give a value of  $V_k$  above one and we have super-persistence—a tendency for the series to cumulatively move above the mean.<sup>37</sup> The key insight of the variance ratio test is that it may be necessary to include a large number of autocorrelations to pick up such mean-reverting behavior. Standard unit root tests or tests of persistence based on short-term

<sup>35</sup> Fama and French (1988), for example, suggested that stock prices contain slowly decaying stationary components that induce negative serial correlation into long-holding-period returns.

<sup>36</sup> In an empirical implementation of the variance ratio test two sources of small sample bias must be corrected for. Cochrane (1988) provides a formula for an unbiased estimator of equation (28').

<sup>37</sup> Lo and MacKinlay (1988) have derived two test statistics that facilitate testing the significance of  $V_k$  (and, in particular, whether  $V_k$  is significantly different from unity or not). The first test statistic is calculated on the assumption that the error term in the real exchange rate process is iid; their second statistic is robust to non-iid errors.

ARMA models (such as those proposed by Campbell and Mankiw (1987)) may fail to capture this mean-reverting behavior.

Huizinga (1987) calculates the variance ratio test (28) for ten dollar bilateral exchange rates for part of the recent floating experience. He does not use the significance tests devised by Lo and MacKinlay, but rather uses the standard  $T^{1/2}$  formula to construct standard errors. Huizinga reports evidence of mean reversion in an economic, or qualitative, sense. By this he means that for all currencies the variance ratio (with a ten-year lag) is numerically below unity (the average across the ten currencies is 0.65 after ten years); however, none of the ratios are statistically different from unity at the 5 percent level. Interestingly, for lags up to about five years the variance ratio for all currencies is above unity, which indicates positive autocorrelation and what we have referred to as super-persistence: the plotted  $V_k$ 's exhibit a hump-shaped profile. One problem, however, with the kind of variance ratio tests implemented by Huizinga is that under the null hypothesis it is assumed the errors in the autoregressive representation are independently and identically distributed (iid). This may not in fact be a good working assumption given the evidence that exchange rates (both real and nominal) display heteroscedasticity.

Glen (1992) has calculated variance ratios, and the corresponding significance tests of Lo and MacKinlay, for a broad selection of U.S. dollar bilateral exchange rates using monthly data. For lags up to and including 32 months Glen finds values of  $V_k$  that are significantly above unity and he is therefore able to reject the random walk null in favor of the alternative of positive serial correlation, or super-persistence. Note, however, that this finding is not supportive of traditional long-run PPP since such super-persistence indicates that a shock to the real exchange rate, rather than pushing it back to its initial value, results in further movements in the same direction. Using an annual data set, however, for the period 1900–87, evidence of mean reversion is found after 4 years, and by 16 years the ratio is 0.433.

Glen's failure to find negative autocorrelation in the monthly data set, which is similar to that used in our multivariate cointegration tests, is rather worrisome since it suggests an important inconsistency in the two sets of results. This inconsistency may reflect the fact that Glen's lag horizon in the monthly data base is simply not long enough to pick up significant mean reversion. Huizinga, as we have noted, required ten-year lags—120 lags with monthly data—to produce mean reversion. In Tables 4 and 5 we therefore present our own estimates of  $V_k$  and the associated significance levels for our currency sets, using lags of 12 through to 120. In terms of the WPI (Table 5), we note that six of the eight

Table 4. *Variance Ratio Statistics—Consumer Prices*

|                | 2    | Lag Length |       |       |       |       |       |       |       |       |       |
|----------------|------|------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                |      | 12         | 24    | 36    | 48    | 60    | 72    | 84    | 96    | 108   | 120   |
| Canada         | 0.96 | 1.14       | 1.39* | 1.17* | 1.82* | 1.78* | 1.66* | 1.52* | 1.64* | 2.02* | 2.19* |
| France         | 0.98 | 1.33*      | 1.73* | 2.03* | 2.09* | 2.14* | 2.10* | 1.97* | 1.63* | 1.31* | 0.88  |
| Germany        | 0.98 | 1.29*      | 1.66* | 1.86* | 1.91* | 2.08* | 2.15* | 2.02* | 1.58* | 1.25  | 0.95  |
| Italy          | 1.03 | 1.49*      | 1.79* | 1.89* | 2.00* | 2.21  | 2.28* | 2.09* | 1.55* | 1.37* | 1.22  |
| Japan          | 1.08 | 1.64*      | 1.86* | 2.03* | 1.97* | 1.79* | 1.83* | 1.76* | 1.35* | 1.77  | 1.24  |
| Netherlands    | 0.89 | 1.29*      | 1.62* | 1.83* | 1.84* | 1.98* | 2.01* | 1.89* | 1.50* | 1.18  | 0.89  |
| Sweden         | 1.04 | 1.19       | 1.61* | 2.14* | 2.44* | 2.70* | 2.85* | 2.91* | 2.61* | 2.01* | 1.23  |
| Switzerland    | 1.03 | 1.26*      | 1.45* | 1.46* | 1.28* | 1.37* | 1.43* | 1.27* | 0.86* | 0.65* | 0.58* |
| United Kingdom | 1.09 | 1.16       | 1.29* | 1.54* | 1.66* | 1.49* | 1.22  | 0.94  | 0.61* | 0.45* | 0.26* |

Notes: The country names in the first column denote the home country currency relative to the U.S. dollar. The numerical entries are the estimates of equation (28') (with an appropriate small sample correction), and an asterisk denotes significance at the 5 percent level, or better, on the basis of the Lo and MacKinlay (1988) Z2 statistic. The latter statistic is a test for deviations of  $V$  from unity, and is robust to non-*iid* errors. The numbers at the top of each column denote the lag length used to construct the variance ratio.

Table 5. *Variance Ratio Statistics—Wholesale Prices*

|                | Lag Length |       |       |       |       |       |       |       |       |       |       |
|----------------|------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                | 2          | 12    | 24    | 36    | 48    | 60    | 72    | 84    | 96    | 108   | 120   |
| Canada         | 0.92       | 0.74* | 0.83  | 0.93  | 0.98  | 1.00  | 0.97  | 0.94  | 0.98  | 1.06  | 1.11  |
| Germany        | 1.00       | 1.08  | 1.34* | 1.40* | 1.39* | 1.53* | 1.56* | 1.39* | 1.00  | 0.83  | 0.75  |
| Italy          | 0.98       | 1.22  | 1.52* | 1.58* | 1.66* | 1.89* | 1.95* | 1.81* | 1.41* | 1.22* | 1.07  |
| Japan          | 0.98       | 1.13  | 1.19  | 1.26* | 1.27* | 1.13  | 1.07  | 0.95  | 0.66* | 0.06* | 0.55* |
| Netherlands    | 0.98       | 1.12  | 1.43* | 1.59* | 1.66* | 1.83* | 1.87* | 1.69* | 1.27* | 1.01  | 0.87  |
| Sweden         | 0.95       | 0.93  | 1.17  | 1.49* | 1.66* | 1.79* | 1.82* | 1.71* | 1.44* | 1.17  | 0.84  |
| Switzerland    | 0.99       | 1.08  | 1.20  | 1.21  | 1.13  | 1.26* | 1.24  | 1.01  | 0.64* | 0.54* | 0.51* |
| United Kingdom | 1.08       | 1.18  | 1.33* | 1.52* | 1.62* | 1.53* | 1.43* | 1.18  | 0.63* | 0.53* | 0.48* |

Notes: See Table 4.

currencies<sup>38</sup> display the hump-shaped pattern noted by Huizinga; Glen's results, therefore, seem to stem from overtruncation of the lag length. Of the six currencies that exhibit mean reversion after ten years, three of the variance ratios are significantly below unity at the 5 percent level (for the United Kingdom, Japan, and Switzerland).<sup>39</sup> There is slightly less evidence of mean reversion for the real exchange rates based on CPIs (Table 4); five of the nine rates display mean-reverting behavior after ten years and two of these are significantly below unity. It is interesting to note the very different patterns in the Japanese yen rate using the CPI and WPI measures. Thus with the WPI, as we have seen, the variance ratio is significantly below unity by lag 120, whereas with the CPI it is still above unity (although not significantly so). This finding confirms Kim's (1990) result for the Japanese yen, noted above.

Abuaf and Jorion (1990) advocate testing for a unit root in real exchange rates by estimating equation (9) directly (rather than imposing a unit root on the testing method as in equation (26')). They propose increasing the efficiency of the estimates by stacking the autoregressive equations for each country into a system (a ZSURE system) and estimating them jointly. Following this method, Abuaf and Jorion show that  $\rho$  lies in the range of 0.98 to 0.99 when monthly data are used. (These results are in fact similar to the constrained results reported in Table 3).<sup>40</sup> Although these point estimates are extremely close to one, they are not exactly one, indicating that there is some evidence of mean reversion.<sup>41</sup> This may therefore suggest that the time-averaging methods of MacDonald and of Flood and Taylor are the important factor in producing a satisfactory PPP result in the context of a panel data set. Instead of moving to pooled average data, however, Abuaf and Jorion move to using a long time span of annual data (1901–72) and report an average slope coefficient of about 0.78. This allows statistical rejection of the null of randomness and defines a half-life of 3.3 years, similar to that implied by the monthly data base.

Diebold, Husted, and Rush (1991) propose a testing method that combines a long time span of data and the method of fractional differencing. The existence of a fractional difference (that is, a value of the difference operator,  $d$ , which lies between zero and unity, rather than being exactly one), in the time-series properties of a variable implies that the variable is stationary. Since this test incorporates a wider range of

<sup>38</sup>The French WPI data stop in 1985 so we have omitted them from our sample.

<sup>39</sup>The German mark is significant at the 10 percent significance level.

<sup>40</sup>With these numbers, it would take between three and five years for a 50 percent over-appreciation of a currency to be cut in half.

<sup>41</sup>See Moore (1993) for a critique of the method used by Abuaf and Jorion. (1990).



stationary processes under the alternative hypothesis it should have greater power to reject the null of a unit root (indeed, Cheung and Lai (1993) have demonstrated that this is the case). Using data on 16 real exchange rates for the gold standard period (for most rates one hundred years of data was available) and a maximum likelihood estimator, they find considerable evidence that  $d$  is significantly below unity; indeed, in some instances the value of  $d$  is insignificantly different from zero. Such evidence is in accord with the view that real rates display mean-reverting behavior (indeed the results where  $d = 0$  are consistent with the level of the rate being stationary—a rather strong result). One telling piece of discussion in the paper is that standard unit root tests were unable to discriminate between a unit root and near unit root behavior even with such long spans of data. This tends to suggest that it is the type of test employed, rather than the observational frequency that is important.

Whitt's (1992) alternative to the standard unit root test is a Bayesian test, attributable to Sims (1988). The Sims-Bayesian approach demonstrates that the prior implicit in the classical unit root approach gives excessive weight to the unit root null.<sup>42</sup> This alternative approach puts a prior on the autoregressive coefficient,  $\rho$ , which spreads the probability,  $\alpha$  (where  $0 < \alpha < 1$ ), of observing  $\rho$  evenly between 0 and 1. The probability of observing a unit root ( $\rho = 1$ ) is  $1 - \alpha$ , which as Whitt notes gives a limited advantage to the unit root hypothesis. Whitt employs two data bases to implement the Sims-Bayesian approach. In particular, the bilateral U.S. dollar rates of the French franc, German mark, Japanese yen, and Swiss franc (based on both CPI and WPI) real exchange rate data for two sample periods: one post-Bretton Woods (monthly observational frequency), the other a period encompassing both Bretton Woods and post-Bretton Woods (annual observational frequency). Using a value for  $\alpha$  of 0.8, Whitt is able to reject the unit root null for each of the real exchange rates considered in each of the two sample periods (although the rejection appears more straightforward for real rate measures using the WPI, perhaps indicating the problems, in terms of the traded/nontraded mix, in more general price series such as the CPI).

## The Real Exchange Rate and Cointegration

As noted in Section I, testing for a unit root in real exchange rates may be interpreted as a rather strict test of PPP. In particular, the condition that forces the real exchange rate to follow a random walk is that *ex ante* real interest rates are equalized across countries. If one uses a sample

<sup>42</sup> Indeed, it gives substantial and excessive weight to values of  $\rho$  above unity; see Sims and Uhlig (1991), p. 8.

period in which long-term capital flows have equalized real rates then this is not a bad assumption to make. However, for the kinds of sample periods that researchers conventionally use this is probably an unreasonable assumption; long-term capital flows do not, even on an average basis, reach a long-run equilibrium. One way of capturing the effect of capital flows on the real exchange rate would be to estimate a version of equation (12'), re-expressed here as a regression equation.<sup>43</sup>

$$q_t = \alpha + \beta(r - r^*)_t + \varphi_t \quad (29)$$

This equation may be derived from (12') by assuming the  $q_{t+k}^*$  term is constant<sup>44</sup> and expectations are formed rationally. This basic relationship has been tested using the Engle-Granger two step method by Meese and Rogoff (1988), Coughlin and Koedijk (1990) and Edison and Pauls (1993) for the recent float. For example, for dollar-mark, dollar-pound, and dollar-yen, over the period February 1974 through December 1985, Meese and Rogoff (1988) test equation (29) using the Engle-Granger two-step method and fail to find any evidence of cointegration. Edison and Pauls also estimate equation (29) for the dollar effective rate, over the period 1974, quarter 3 through to 1990, quarter 4. Using a variety of different proxies for expected inflation they cannot reject the null of no cointegration for the trade-weighted value of the dollar and the bilateral U.S. dollar rates of the mark, yen, pound sterling, and Canadian dollar. Furthermore, they also fail to reject the null of no cointegration when the potential nonconstancy of  $E_t q_{t+k}$  is allowed for (by assuming that  $E_t q_{t+k}$  is a function of the cumulated current account and introducing this as an extra explanatory variable). The only paper in this genre to find some evidence of cointegration for equation (29) is that of Coughlin and Koedijk (1990), who report cointegration between the German mark-U.S. dollar rate and the real interest rate differential.

There are, however, at least two major problems with this kind of test, each of which may explain the failure to reject the null of noncointegration. First, we know from the various tests discussed above that the real exchange rate is a stationary process and there is also evidence (noted below) to suggest that the real interest differential is stationary. What sense then does it make to model the real rate differential as an I(1) process (which is what is required for a cointegration test based on (29))?

<sup>43</sup> One can think of this relationship as extending the cointegrating set. Thus to the extent that unit root tests on  $q$  indicate an absence of cointegration, adding additional appropriate nonstationary variables to the cointegrating set may produce a cointegrating relationship.

<sup>44</sup> The plausibility of this assumption is doubtful and is discussed further below.

Second, all of the above-noted tests have involved the Engle-Granger two-step method; it is now widely accepted that this method is not a particularly powerful test of the null hypothesis of noncointegration when the null is in fact false.

### The Real Interest Rate/Exchange Rate Link— Some Further Evidence

Before closing this section it is worth discussing some other research, which examines the real exchange rate/real interest rate link using methods other than cointegration. Meese and Rogoff (1988) treat equation (29) as a standard regression equation and, assuming that both  $q$  and  $r - r^*$  are nonstationary (although, as we have noted, this is not uncontroversial), regress the first difference of the real rate on the first difference of the real interest differential.<sup>45</sup> Not only are Meese and Rogoff unable to find a value of  $\beta$  that is significantly above unity (for the three currencies noted above), they cannot reject the hypothesis that  $\beta$  is insignificantly different from zero. They also try “sharpening up” the specification by modeling the equilibrium real rate (assumed equal to  $q_{t+k}^*$ ) using home and foreign trade balances. However, these extended regressions do not result in any satisfactory improvement in the estimates.

A conclusion similar to that proposed by Meese and Rogoff is given by Campbell and Clarida (1987). Using an unobserved components model they demonstrate that the majority of movements in the real exchange rate (at least 79 percent) are driven by movements in the permanent component of the real rate (that is,  $q_{t+k}^*$  the long-run component) and a very small component is due to real interest differentials (this is shown never to exceed 9 percent). Campbell and Clarida do, however, find that the implied value of  $\beta$  is greater than one in absolute terms, implying that the real exchange rate is more volatile than the real interest differential by a factor of about ten.

Baxter (1994) forcefully argues that the failure of studies like Meese and Rogoff and Campbell and Clarida to uncover any worthwhile relationships between real exchange rates and real interest differentials is due to the particular interpretation of the relationship. For example, by using

<sup>45</sup>The real interest rates are defined on an ex post basis; that is by subtracting the ex post realized inflation rate from the nominal interest rate. Such an approach introduces a moving average error structure into the estimated equation and the OLS standard errors are corrected using a Generalized Method of Moments correction (which also corrects for heteroscedasticity).

a first-difference transformation, Meese and Rogoff presume that the relationship between the variables relates to the permanent elements (that is, assuming a unit root representation presupposes that any change in each of the variables is a permanent change). As Baxter notes, however, the first-difference operator, although removing the unit root from an economic time series, also removes most of the other low-frequency information. Moreover, she demonstrates that the key correlation (or prediction from the sticky price model) between  $q$  and  $r - r$  is between the *temporary* components of the real rate and the real differential.<sup>46</sup> Using univariate and multivariate Beveridge and Nelson (1981) decompositions of the real exchange rate, Baxter demonstrates statistically significant values for  $\beta$ , especially when the multivariate decomposition is employed.

## V. Concluding Comments

In this paper we have surveyed the recent empirical literature on the existence of a long-run exchange rate relationship. This literature has had something of a symbiotic relationship with recent developments in the time-series literature and, in particular, the literature on cointegration. In summary, the research overviewed in this paper, and our own empirical results, strongly suggest the existence of "some form" of long-run exchange rate relationship.<sup>47</sup> The form of this long-run relationship may be understood by detailing our main findings. In summarizing our findings it is useful to consider two distinct periods: the recent floating period and sample periods that extend back before the start of the float.

For the recent floating period there is now considerable evidence supportive of a long-run relationship between relative prices and exchange rates, in the sense that these variables are cointegrated. However, the long-run relationship does not conform to traditional PPP since the hypothesis of proportionality is often rejected (indeed the symmetry hypothesis is also frequently rejected). There are essentially three ways to explain this finding. First, and as we noted in Section III, there may be measurement errors and traded/nontraded biases that generate this result, although it is hard to imagine that the gross violations observed in the data can be wholly attributed to these kinds of factors. Second, the

<sup>46</sup> This framework makes clear that it does not make sense to test for cointegration between the level of a real exchange rate and the real interest differential.

<sup>47</sup> This evidence seems strongest for relationships based on the wholesale price measure, probably because the traded good component is much larger than for the CPI.

econometric methods used may produce coefficient biases that can be substantial in small samples; some evidence reported in this paper suggests that this is indeed the case. The third interpretation is that there have been real disturbances and capital movements during the recent float which upset the proportionality relationship. We consider this interpretation in more detail below. One interesting result from the cointegration tests is that the adjustment to equilibrium is relatively fast (where we emphasize that the equilibrium does not exhibit proportionality) and has, with monthly data, a half-life of about 3 years.

In contrast, using standard (i.e., Dickey-Fuller) unit root tests to detect mean reversion in real exchange rates for the recent float proves to be a disappointing exercise, since the null hypothesis of a unit root cannot be rejected with such tests. However, it is widely accepted that the standard tests have relatively low power against a broader range of stationary alternatives. Indeed, alternative tests for unit roots in real exchange rate are more successful in rejecting the null hypothesis. However, these tests suggest that the degree of mean reversion is painfully slow. For example, the variance ratio tests reported in this paper indicate that significant mean reversion only starts to occur after eight years. One way of substantially reducing this number is to consider real exchange rates as a panel and test for mean reversion across the panel. This kind of test produces a half-life of deviations from PPP of about four years.

A similar half-life to that reported in panel regressions has been found in individual real exchange rates when the data span is extended prior to the recent float (in particular, using about 100 years of annual data). Long time spans of data also produce significant and sensible (in the sense that proportionality cannot be rejected) long-run relationships between nominal exchange rates and relative prices. These long spans of data give the real exchange rate a greater opportunity to cross the mean after a real disturbance, an opportunity that is considerably reduced by the relatively short sample available from the recent float. However, it is undoubtedly still of interest to model individual exchange rates for the recent floating period.

Given this, how may one tie down a sensible equilibrium to which the exchange rate adjusts? In the context of the research discussed in this paper, we believe that a sensible long-run relationship can be defined, and a relatively rapid adjustment to that equilibrium established, by recognizing the distinction between true and statistical equilibrium noted in Section I. The true equilibrium is one that accords to what most economic models would posit as a long-run solution. Thus, the sticky-price monetary model of Dornbusch (1976) and the flex-price model of Frenkel (1976) and Hodrick (1978) both have absolute PPP as their

long-run solution.<sup>48</sup> The statistical equilibrium is simply the one that is captured by the particular econometric or statistical technique used to estimate the long-run exchange rate. Thus, the Johansen method captures the mean-reverting properties of the nominal exchange rate with respect to relative prices for the recent floating experience, but it fails to capture the symmetry and homogeneity restrictions required by the true long-run equilibrium. The two concepts may, however, be reconciled by an explicit recognition of the distinction when undertaking an actual estimation. Thus, in estimating PPP for the recent floating experience a suitable estimating equation should be derived from an equilibrium condition that conforms to the period studied, rather than from a true equilibrium condition. Following MacDonald and Marsh (1994b), we would argue that the relevant equilibrium for a period such as the recent floating period is the balance of payments equilibrium condition introduced in Section 11.

With freely floating exchange rates, the exchange rate should move to ensure that the sum of the current and capital accounts, or the change in reserves, is equal to zero. However, most theoretical models of exchange rate determination would define a true long-run equilibrium as one in which the current account equals zero (and by implication net capital flows are zero). It is our contention that for a sample period such as the recent float, net capital flows will not go to zero and, therefore, they should be explicitly recognized in modeling the measure of the long-run exchange rate currently adopted in the literature. Of particular importance in this regard are long-term capital flows, which reflect productivity and thrift factors and also expected inflation, and which imply that a relationship that conditions exchange rates solely on relative prices will not tell the full story. MacDonald and Marsh (1994b) have exploited this type of distinction for the recent float for the U.S. dollar bilateral rates of the German mark, Japanese yen, and U.K. pound. They find that conditioning the exchange rate on relative prices and long-term bond yields produces cointegration and, crucially, a failure to reject the proportionality and symmetry of the exchange rates with respect to relative prices. Furthermore, adjustment to this equilibrium is rapid. MacDonald and Marsh then use the estimated long-run relationships to produce short- to medium-run dynamic exchange rate models that perform significantly better (in a statistical sense) than a simple random walk in terms of their out-of-sample forecasting accuracy. We believe that this kind of approach merits further attention, especially when a researcher is limited to data from the recent floating experience.

<sup>48</sup> In the flex-price monetary model, the long run holds continuously.

MacDonald (1995a) has fleshed out the links between exchange rates and long bond yields discussed in MacDonald and Marsh (1994b). In particular, he explicitly models the effects of, *inter alia*, productivity differences on the long-run relationships for the nominal effective exchange rates of the German mark, Japanese yen, and U.S. dollar, for the recent floating period. In addition to finding statistically significant long-run relationships, rapid adjustment occurs with respect to the equilibrium rates.<sup>49</sup>

The recent success in modeling both short- and long-run exchange rates (see also MacDonald and Taylor (1993 and 1994) is encouraging and should serve as a healthy counterpoint to the recent move away from fundamentals and toward nonfundamental explanations of exchange rate behavior, such as chartism and market microstructure.

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<sup>49</sup>The average adjustment per quarter across the three currencies is about 10 percent, giving a half-life of approximately a year and a half.

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